

Summary :

- ❑ Concept of Gradient of a scalar field, Divergence and Curl of a vector field
- ❑ Line, surface and volume integral of a vector field
- ❑ Fundamental theorem of these integrals:

- The fundamental Theorem for Gradient:
$$\int_a^b \nabla T \cdot d\mathbf{l} = T(b) - T(a)$$

- The fundamental Theorem for Divergence (**Gauss's theorem**):
$$\int_{Vol} (\nabla \cdot \mathbf{V}) d\tau = \oint_{Surf} \mathbf{V} \cdot d\mathbf{a}$$

- The fundamental Theorem for Curl (**Stokes' theorem**):
$$\int_{Surf} (\nabla \times \mathbf{V}) \cdot d\mathbf{a} = \oint_{Path} \mathbf{V} \cdot d\mathbf{l}$$

- Coordinate systems: Spherical polar, and cylindrical

Essential concepts :

1. The Helmholtz theorem

A vector field $\mathbf{F}$ in electrodynamics can be completely determined if:

- (i) The divergence $\nabla \cdot \mathbf{F}$ is known
- (ii) The curl $\nabla \times \mathbf{F}$ is known
- (iii) If the field goes to zero at infinity.

Essential concepts :

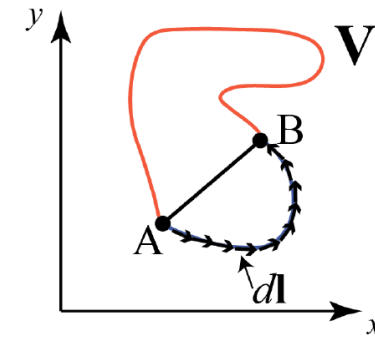
2. Scalar Potential:

If the curl of a vector field $\mathbf{F}$ is zero, that is, if $\nabla \times \mathbf{F} = 0$ everywhere, then:

$$\int_{\text{Surf}} (\nabla \times \mathbf{F}) \cdot d\mathbf{a} = \oint_{\text{Path}} \mathbf{F} \cdot d\mathbf{l} = 0$$

$\oint \mathbf{F} \cdot d\mathbf{l} = 0$ for any closed loop.

$\int_a^b \mathbf{F} \cdot d\mathbf{l}$ is independent of path.



$\mathbf{F}$ is the gradient of a scalar function: $\mathbf{F} = -\nabla V$ as $\nabla \times (\nabla V) = 0$

V is called Scalar potential

- The minus sign is purely conventional.
- The scalar potential not unique. A constant can be added to V without affecting the gradient: $\nabla V = \nabla(V + a)$, since the gradient of a constant is zero.

Essential concepts :

3. Vector Potential:

If the divergence of a vector field $\mathbf{F}$ is zero, that is, if $\nabla \cdot \mathbf{F} = 0$ everywhere, then:

$$\int_{Vol} (\nabla \cdot \mathbf{F}) d\tau = \oint_{Surf} \mathbf{F} \cdot d\mathbf{a} = 0$$

$\oint \mathbf{F} \cdot d\mathbf{a} = 0$ for any closed surface.

$\int \mathbf{F} \cdot d\mathbf{a}$ is independent of surface.

$\mathbf{F}$ is the gradient of a vector function: $\mathbf{F} = \nabla \times \mathbf{A}$ as $\nabla \cdot (\nabla \times \mathbf{A}) = 0$

$\mathbf{A}$ is called Vector potential

Note that the vector potential is not unique and function can be added to $\mathbf{A}$ w in the form of gradient of a scalar function, without affecting the curl, since the curl of a gradient is zero.

$$\nabla \times \mathbf{A} = \nabla \times (\mathbf{A} + \nabla\varphi) \quad \text{as} \quad \nabla \times (\nabla\varphi) = 0$$

Interesting observations

Q: Find the divergence of the function $\mathbf{v}(r) = \frac{\hat{r}}{r^2}$

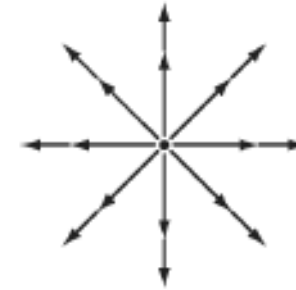
$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{1}{r^2} \right) = \frac{1}{r^2} \frac{\partial}{\partial r} (1) = 0.$$

$$\int_{Vol} (\nabla \cdot \mathbf{V}) d\tau = 0$$

The surface integral over a close surface :

$$\int_{Vol} (\nabla \cdot \mathbf{F}) d\tau = \oint_{Surf} \mathbf{F} \cdot d\mathbf{a}$$

Excluding $\mathbf{r} = 0$ solves the problem !!!!



$$\begin{aligned} \oint \mathbf{v} \cdot d\mathbf{a} &= \int \left(\frac{1}{R^2} \hat{\mathbf{r}} \right) \cdot (R^2 \sin \theta d\theta d\phi \hat{\mathbf{r}}) \\ &= \left(\int_0^\pi \sin \theta d\theta \right) \left(\int_0^{2\pi} d\phi \right) = 4\pi. \end{aligned}$$

δ - Function

$$\delta(x) = \begin{cases} 0, & \text{if } x \neq 0 \\ \infty, & \text{if } x = 0 \end{cases}$$

Delta function:

We Define $\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta(\mathbf{r}) = 4\pi \delta(\mathbf{r} - \mathbf{r}')$

$$\delta(x) = \begin{cases} 0, & \text{if } x \neq 0 \\ \infty, & \text{if } x = 0 \end{cases}$$

$$\int_a^b \delta(\mathbf{r}) d\mathbf{r} = 1 \quad \& \quad \int_a^b f(\mathbf{r}) \delta(\mathbf{r} - \mathbf{r}') d\mathbf{r} = f(\mathbf{r}')$$

For the function $\mathbf{v}(r) = \frac{\hat{\mathbf{r}}}{r^2}$, $\nabla \cdot \mathbf{v}(r) = 4\pi\delta(r)$

Hence

$$\int_{vol} \nabla \cdot \mathbf{v}(r) d\tau = \int_{vol} 4\pi\delta(r) d\tau = 4\pi$$

$$\oint_{surface} \mathbf{v}(r) \cdot d\mathbf{a} = \oint_{surface} \left(\frac{\hat{\mathbf{r}}}{R^2} \right) \cdot (R^2 \sin \theta d\theta d\varphi \hat{\mathbf{r}}) = 4\pi$$

$$\int_{vol} \nabla \cdot \mathbf{v}(r) d\tau = \oint_{surface} \mathbf{v}(r) \cdot d\mathbf{a}$$

Electrostatics

Electric Field

Coulomb's Law

Force on a test charge Q due to a single point charge q is:

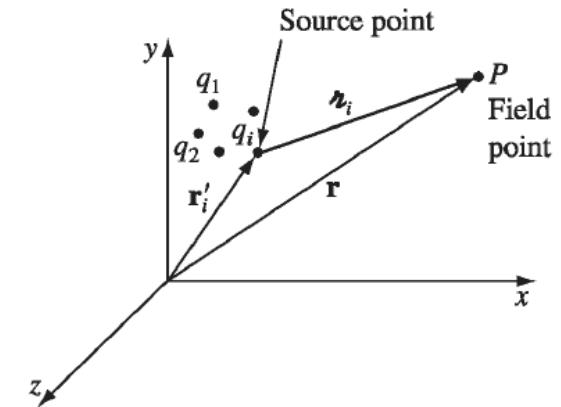
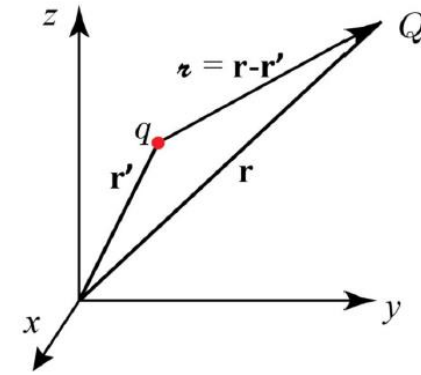
$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{\mathbf{r}} \quad \epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{N} \cdot \text{m}^2} \quad \text{Permittivity of free space}$$

$$\mathbf{F} = \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \right) Q = \vec{E}Q, \quad \text{Where } \vec{E} = \left(\frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \right) \text{ is called the electric field}$$

If we have *several* point charges $q_1, q_2, \dots, q_n$, at distances $r_1, r_2, \dots, r_n$ from Q , the total force on Q is evidently

$$\begin{aligned} \mathbf{F} &= \mathbf{F}_1 + \mathbf{F}_2 + \dots = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 Q}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_2 Q}{r_2^2} \hat{\mathbf{r}}_2 + \dots \right) \\ &= \frac{Q}{4\pi\epsilon_0} \left(\frac{q_1}{r_1^2} \hat{\mathbf{r}}_1 + \frac{q_2}{r_2^2} \hat{\mathbf{r}}_2 + \frac{q_3}{r_3^2} \hat{\mathbf{r}}_3 + \dots \right) = QE, \end{aligned}$$

$$\mathbf{E}(\mathbf{r}) \equiv \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{\mathbf{r}}_i.$$



Electrostatics

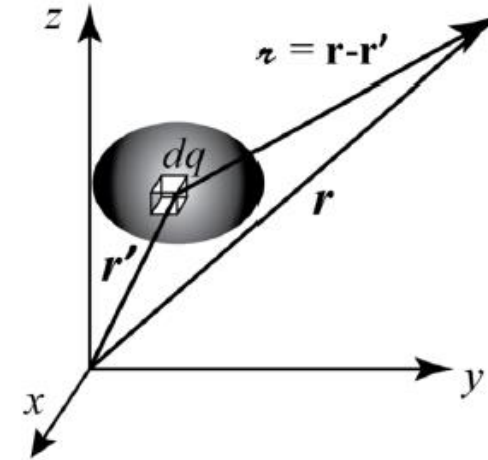
Electric field due to a continuous charge distribution is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{\mathbf{r}}$$

For a line charge $dq = \lambda(\mathbf{r}')dl$

For a surface charge $dq = \sigma(\mathbf{r}')da$

For a volume charge $dq = \rho(\mathbf{r}')d\tau$



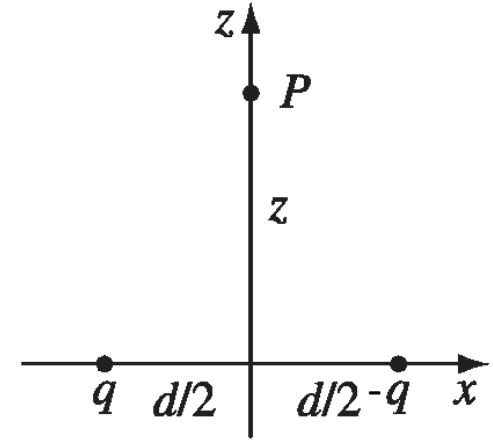
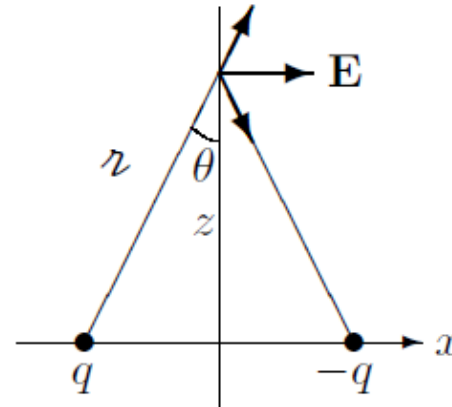
We know how to calculate electric fields due a charge distribution, using Coulomb's law.

We'll now explore some tricks for calculating the field more efficiently.

Problem: Find the electric field (magnitude and direction) a distance Z above the midpoint between equal and opposite charges $(\pm q)$, d distanced apart

“vertical” components cancel,

$$\begin{aligned}\mathbf{E} &= \frac{1}{4\pi\epsilon_0} 2 \frac{q}{r^2} \sin \theta \hat{\mathbf{x}}, \\ &= \frac{1}{4\pi\epsilon_0} \frac{qd}{\left(z^2 + \left(\frac{d}{2}\right)^2\right)^{3/2}} \hat{\mathbf{x}}.\end{aligned}$$



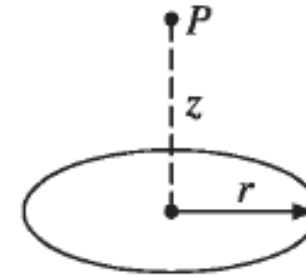
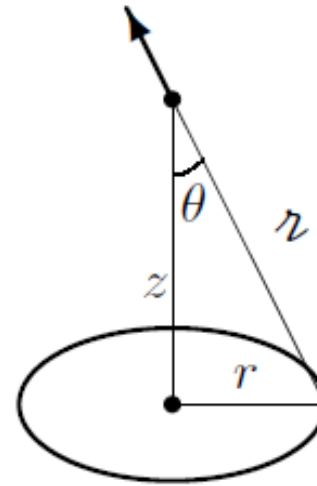
Problem: Find the electric field a distance Z above the center of a circular loop of radius r that carries a uniform line charge λ .

“Horizontal” components cancel,

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \left\{ \int \frac{\lambda dl}{r^2} \cos \theta \right\} \hat{\mathbf{z}}$$

$$r^2 = r^2 + z^2, \cos \theta = \frac{z}{r} \text{ (both constants)}$$

$$\int dl = 2\pi r$$



$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda(2\pi r)z}{(r^2 + z^2)^{3/2}} \hat{\mathbf{z}}.$$

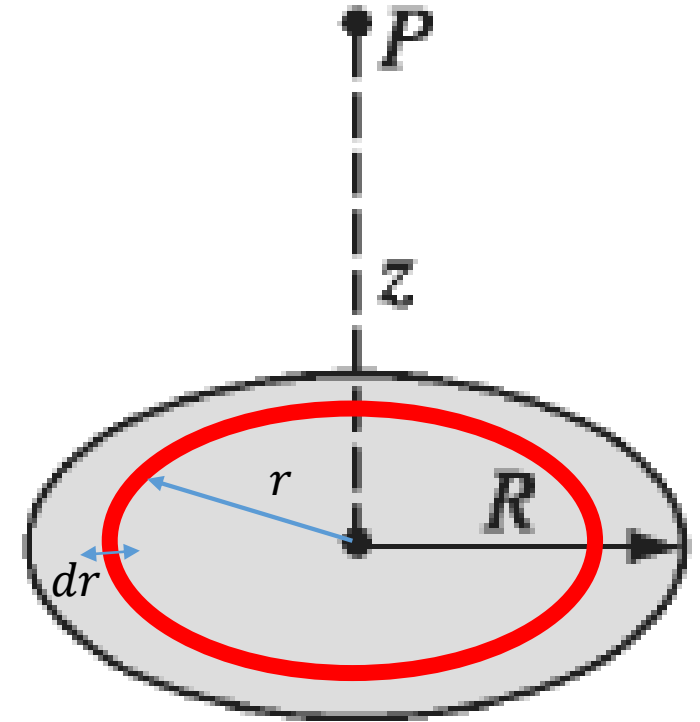
Problem: Find the electric field a distance z above the center of a flat circular disk of radius R that carries a uniform surface charge σ . What happens if $R \rightarrow \infty$? Also if $z \gg R$

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda(2\pi r)z}{(r^2 + z^2)^{3/2}} \hat{\mathbf{z}}.$$

$$\sigma \cdot 2\pi r \cdot dr = \lambda \cdot 2\pi r, \text{ so } \lambda = \sigma dr$$

$$E_{\text{ring}} = \frac{1}{4\pi\epsilon_0} \frac{(\sigma dr)2\pi rz}{(r^2 + z^2)^{3/2}};$$

$$E_{\text{disk}} = \frac{1}{4\pi\epsilon_0} 2\pi\sigma z \int_0^R \frac{r}{(r^2 + z^2)^{3/2}} dr = \frac{1}{4\pi\epsilon_0} 2\pi\sigma z \left[\frac{1}{z} - \frac{1}{\sqrt{R^2 + z^2}} \right] \hat{\mathbf{z}}$$



If $R \rightarrow \infty$, then the second term goes to 0

$$\mathbf{E}_{\text{plane}} = \frac{1}{4\pi\epsilon_0} 2\pi\sigma \hat{\mathbf{z}} = \boxed{\frac{\sigma}{2\epsilon_0} \hat{\mathbf{z}}}.$$

$$\text{For } z \gg R, \frac{1}{\sqrt{R^2 + z^2}} = \frac{1}{z} \left(1 + \frac{R^2}{z^2} \right)^{-1/2} \approx \frac{1}{z} \left(1 - \frac{1}{2} \frac{R^2}{z^2} \right)$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{2\pi R^2 \sigma}{2z^2} = \frac{1}{4\pi\epsilon_0} \frac{Q}{z^2} \quad \text{where } Q = \pi R^2 \sigma.$$

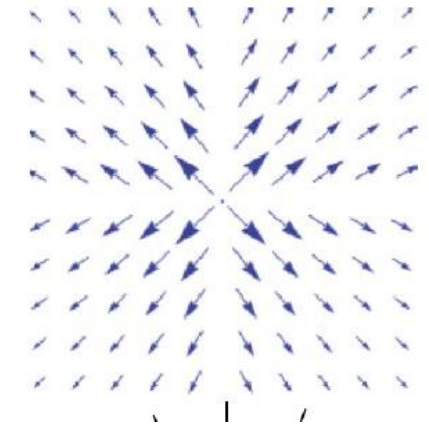
Electrostatics

Electric Flux and Gauss's Law:

Electric field $\mathbf{E}(\mathbf{r})$ due to a single point charge q at origin is:

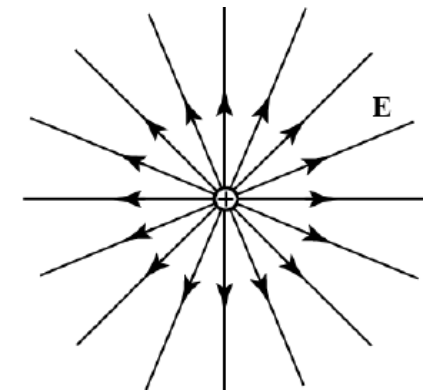
$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

Electric field is a vector field. One way to represent a vector field is by drawing a vectors of given magnitude and directions.



Another way to represent a vector field is by drawing the field lines:

- (i) Field lines emanate from the positive charge and end up on the negative charge or go up to infinity.
- (ii) The density of the field lines is proportional to the strength of the field.



Electrostatics

Electric Flux and Gauss's Law:

Electric field $\mathbf{E}(\mathbf{r})$ due to a single point charge q at origin is:

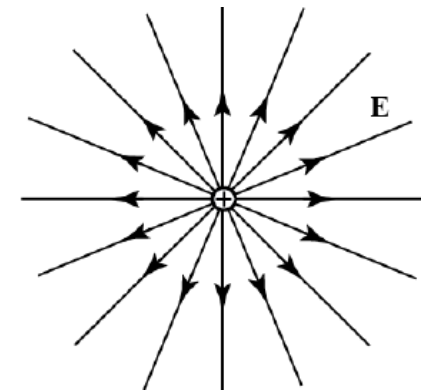
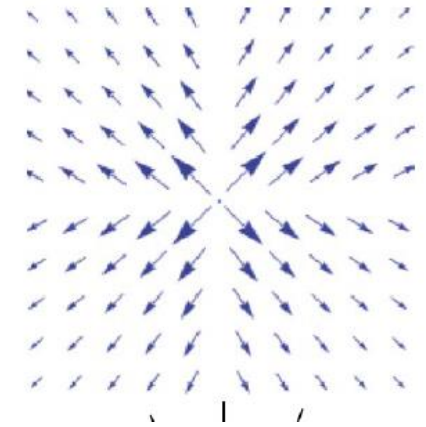
$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

The electric flux is defined as:

$$\Phi_E = \int \mathbf{E} \cdot d\mathbf{a}$$

$\mathbf{E} \cdot d\mathbf{a}$ is proportional the number of field lines passing through an area element $d\mathbf{a}$

When the area $d\mathbf{a}$ is perpendicular to the field $\mathbf{E}$ the dot product is zero.



Electrostatics

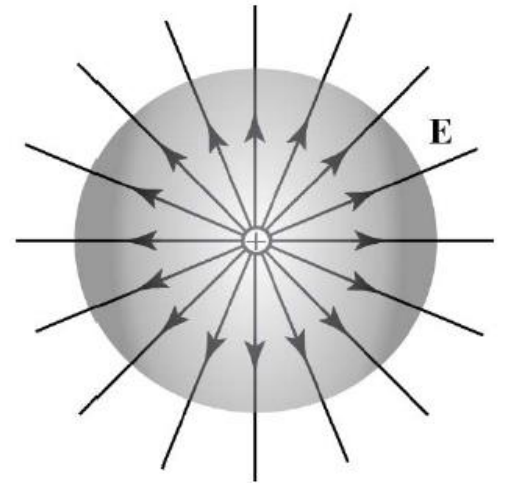
Electric Flux and Gauss's Law:

Electric field $\mathbf{E}(\mathbf{r})$ due to a single point charge q at origin is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

The electric flux due to a point charge q at origin through a spherical shell of radius R .

$$\Phi_E = \oint_{surf} \mathbf{E} \cdot d\mathbf{a} = \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} \hat{\mathbf{r}} \cdot R^2 \sin\theta d\theta d\phi \hat{\mathbf{r}} = \frac{q}{\epsilon_0}$$



Note: Φ_E not depend on the radius

Electrostatics

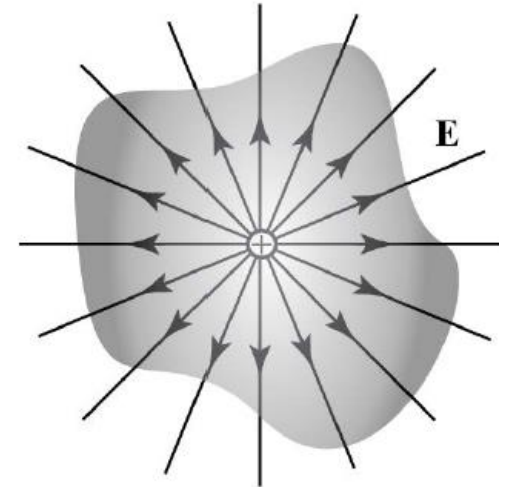
Electric Flux and Gauss's Law:

Electric field $\mathbf{E}(\mathbf{r})$ due to a single point charge q at origin is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

The electric flux due to a point charge q at origin through any enclosing closed surface is

$$\Phi_E = \oint_{surf} \mathbf{E} \cdot d\mathbf{a} = \frac{q}{\epsilon_0}$$



Electrostatics

Electric Flux and Gauss's Law:

Electric field $\mathbf{E}(\mathbf{r})$ due to a single point charge q at origin is:

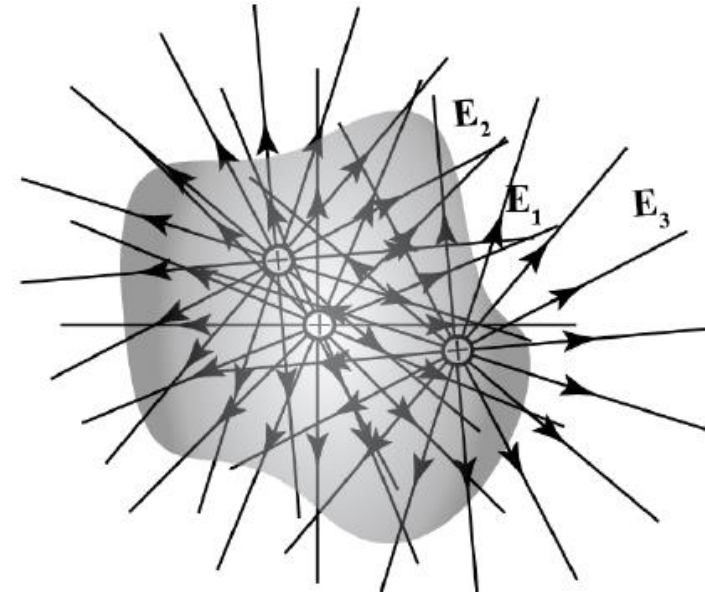
$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

The electric flux due to a collection of point charges through any enclosing closed surface.

$$\begin{aligned}\Phi_E &= \oint_{surf} \mathbf{E} \cdot d\mathbf{a} = ?? \\ &= \oint_{surf} \left(\sum_{i=1}^n \mathbf{E}_i \right) \cdot d\mathbf{a} = \sum_{i=1}^n \oint_{surf} \mathbf{E}_i \cdot d\mathbf{a} = \sum_{i=1}^n \frac{q_i}{\epsilon_0}\end{aligned}$$

$$\oint_{surf} \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0}$$

This is the Gauss's law in the integral form:
The flux through a surface is equal to the total charge enclosed by the surface divided by ϵ_0



Electric Flux and Gauss's Law:

$$\oint_{surf} \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0}$$

This is the Gauss's law in integral form.

$$\text{Or, } \int_{Vol} (\nabla \cdot \mathbf{E}) d\tau = \frac{Q_{enc}}{\epsilon_0}$$

Using divergence
theorem

$$\int_{Vol} (\nabla \cdot \mathbf{E}) d\tau = \oint_{Surf} \mathbf{E} \cdot d\mathbf{a}$$

NOTE: Gauss's law doesn't provide any extra information other than that of Coulomb's Law provides. The importance of Gauss's law is that it makes calculating electric field much simpler and provide a deeper understanding of the field itself.

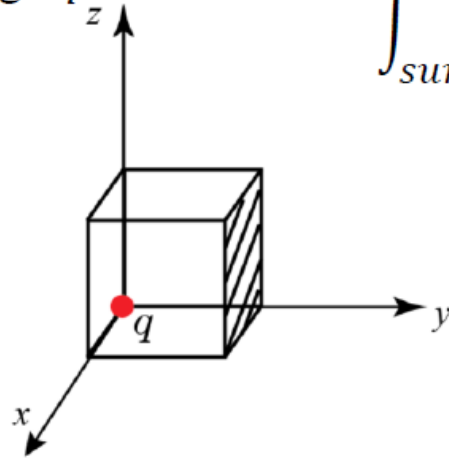
Electric Flux and Gauss's Law:

$$\oint_{surf} \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0}$$

This is the Gauss's law in integral form.

Q: (Griffiths: Ex 2.10): What is the flux through the shaded face of the cube due to the charge q at the corner

$$\int_{surf} \mathbf{E} \cdot d\mathbf{a} ??$$



Gauss's Law from Coulomb's Law:

If Coulomb's Law and Gauss's law have the same information content, can we derive Gauss's law from Coulomb's law?

Coulomb's law gives the electric field due to a volume charge $\rho(\mathbf{r}')$

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\hat{\mathbf{r}}}{r^2} \rho(\mathbf{r}') d\tau$$

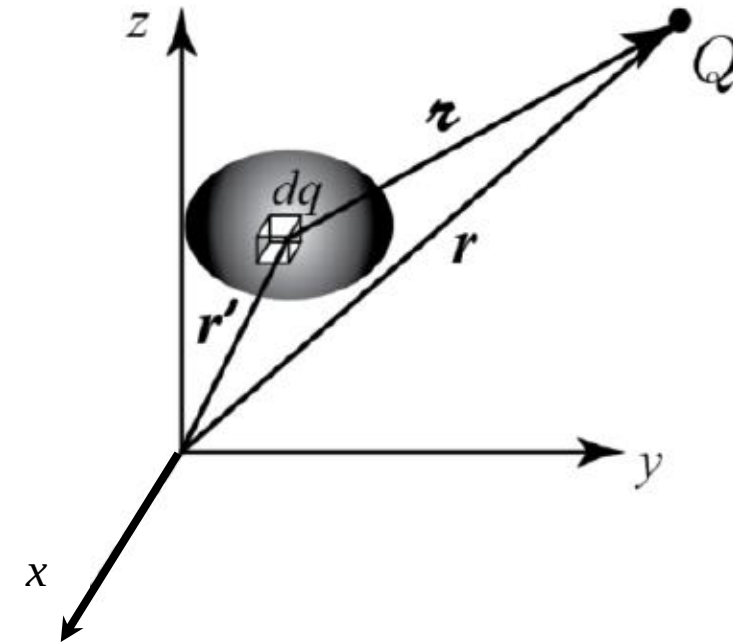
Take the divergence of both sides of the equation

$$\nabla \cdot \mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) \rho(\mathbf{r}') d\tau$$

We have: $\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta(\mathbf{r}) = 4\pi \delta(\mathbf{r} - \mathbf{r}')$

$$\text{Therefore, } \nabla \cdot \mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int 4\pi \delta(\mathbf{r} - \mathbf{r}') \rho(\mathbf{r}') d\tau = \frac{\rho(\mathbf{r})}{\epsilon_0}$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$



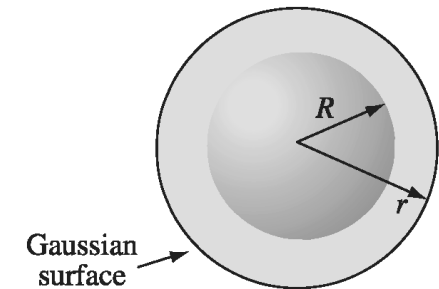
The divergence of electric field is equal to the charge density divided by ϵ_0

Application of Gauss's Law

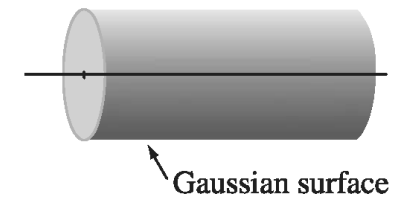
When symmetry permits, the easiest way of computing electric fields is by using Gauss's Law.

1. Construct Gauss surface around the charge distribution
2. Calculate the surface integral.
3. Equate the result with (enclosed charges $)/\epsilon_0$

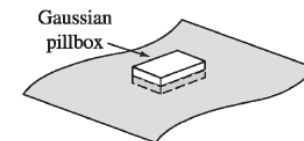
A. *Spherical symmetry.* Make a Gaussian surface a concentric sphere.



B. *Cylindrical symmetry.* Make a Gaussian surface a coaxial cylinder



C. *Plane symmetry.* Use a Gaussian "pillbox" that straddles the surface.



Application of Gauss's Law

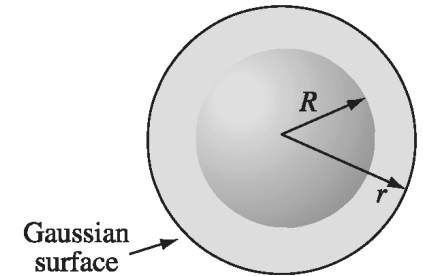
Problem: Find the field outside a uniformly charged solid sphere of radius R and total charge q .

Imagine a spherical surface at radius $r > R$ (Gaussian surface).

Gauss's law says that $\oint_S \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}},$

$$\int_S \mathbf{E} \cdot d\mathbf{a} = \int_S |\mathbf{E}| da, = |\mathbf{E}| \int_S da = |\mathbf{E}| 4\pi r^2$$

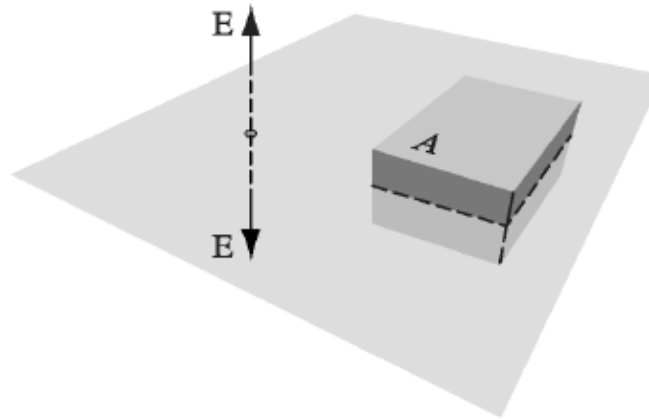
$$|\mathbf{E}| 4\pi r^2 = \frac{1}{\epsilon_0} q, \quad \Rightarrow \quad \mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}.$$



Application of Gauss's Law

Problem: An infinite plane carries a uniform surface charge σ . Find its electric field.

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$



$$\int \mathbf{E} \cdot d\mathbf{a} = 2A|\mathbf{E}|, \quad Q_{\text{enc}} = \sigma A,$$

$$2A|\mathbf{E}| = \frac{1}{\epsilon_0} \sigma A,$$

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}},$$

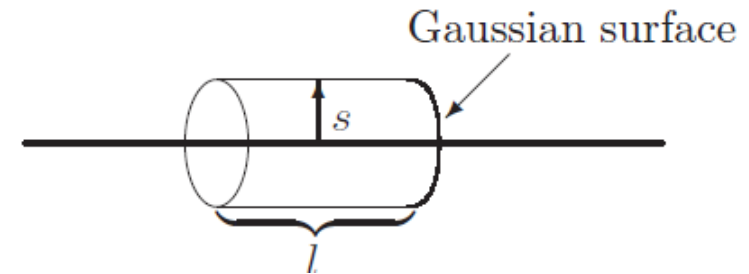
Application of Gauss's Law

Problem: Find the electric field a distance s from an infinitely long straight wire that carries a uniform line charge.

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}}$$

$$\oint \mathbf{E} \cdot d\mathbf{a} = E \cdot 2\pi s \cdot l \qquad \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{1}{\epsilon_0} \lambda l.$$

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{\mathbf{s}}$$



Problem 2.11 Use Gauss's law to find the electric field inside and outside a spherical shell of radius R that carries a uniform surface charge density σ . Compare your answer to Prob. 2.7.

Problem 2.12 Use Gauss's law to find the electric field inside a uniformly charged solid sphere (charge density ρ). Compare your answer to Prob. 2.8.

Problem 2.13 Find the electric field a distance s from an infinitely long straight wire that carries a uniform line charge λ . Compare Eq. 2.9.

Problem 2.14 Find the electric field inside a sphere that carries a charge density proportional to the distance from the origin, $\rho = kr$, for some constant k . [Hint: This charge density is *not* uniform, and you must *integrate* to get the enclosed charge.]

Problem 2.15 A thick spherical shell carries charge density

$$\rho = \frac{k}{r^2} \quad (a \leq r \leq b)$$

(Fig. 2.25). Find the electric field in the three regions: (i) $r < a$, (ii) $a < r < b$, (iii) $r > b$. Plot $|\mathbf{E}|$ as a function of r , for the case $b = 2a$.

Problem 2.16 A long coaxial cable (Fig. 2.26) carries a uniform *volume* charge density ρ on the inner cylinder (radius a), and a uniform *surface* charge density on the outer cylindrical shell (radius b). This surface charge is negative and is of just the right magnitude that the cable as a whole is electrically neutral. Find the electric field in each of the three regions: (i) inside the inner cylinder ($s < a$), (ii) between the cylinders ($a < s < b$), (iii) outside the cable ($s > b$). Plot $|\mathbf{E}|$ as a function of s .

Curl of the Electric Field:

Let's take the simplest electric field:

Electric field due to a single point charge q is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

We need to find the curl of it

$$\nabla \times \mathbf{E}(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \nabla \times \left(\frac{1}{r^2} \hat{\mathbf{r}} \right)$$

Take the area integral

$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \int_{surf} \nabla \times \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) \cdot d\mathbf{a}$$

Use Stokes's theorem

$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \oint \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) \cdot d\mathbf{l} = \frac{q}{4\pi\epsilon_0} \oint \frac{1}{r^2} dr$$

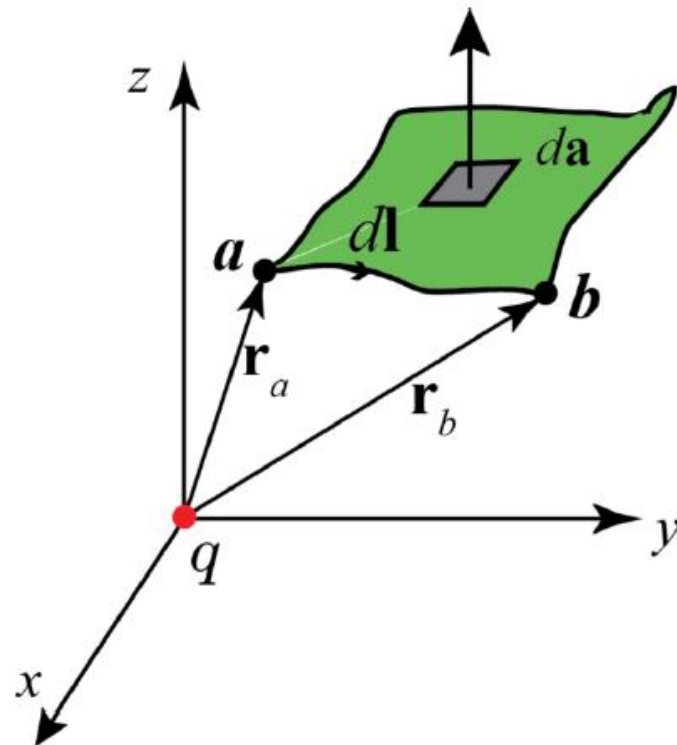
since

$$d\mathbf{l} = dr\hat{\mathbf{r}} + rd\theta\hat{\boldsymbol{\theta}} + r\sin\theta d\phi\hat{\boldsymbol{\phi}}$$

$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \times \frac{1}{r} \Big|_{r_a}^{r_a} = 0$$

implies

$$\nabla \times \mathbf{E} = 0$$



Electric Potential:

Recall: If the curl of a vector field $\mathbf{F}$ is zero, that is, if $\nabla \times \mathbf{F} = 0$ everywhere, then:

(1) $\int_a^b \mathbf{F} \cdot d\mathbf{l}$ is independent of path.

(2) $\oint \mathbf{F} \cdot d\mathbf{l} = 0$ for any closed loop.

(3) $\mathbf{F}$ is the gradient of a scalar function: $\mathbf{F} = -\nabla V$

This is because of Stokes' theorem

$$\int_{\text{Surf}} (\nabla \times \mathbf{F}) \cdot d\mathbf{a} = \oint_{\text{Path}} \mathbf{F} \cdot d\mathbf{l}$$

• This is because Curl of a gradient is zero $\nabla \times (\nabla V) = 0$

The curl of Electric field $\mathbf{E}$ is zero, that is, $\nabla \times \mathbf{E} = 0$ everywhere. Therefore:

$\oint \mathbf{E} \cdot d\mathbf{l} = 0$ for any closed loop.



$\int_a^b \mathbf{E} \cdot d\mathbf{l}$ is independent of path.

$\mathbf{E}$ is the gradient of a scalar function:

$$\mathbf{E} = -\nabla V$$

V is called the electric potential. It is a scalar quantity, the gradient of which is equal to the electric field

How to obtain/write the electric potential in terms of $\mathbf{E}$??

Electric Potential:

Since $\nabla \times \mathbf{E} = 0$ everywhere, implies $\int_a^b \mathbf{E} \cdot d\mathbf{l}$ is independent of path(depends only on the end points)

$$\int_a^b \mathbf{E} \cdot d\mathbf{l} = V(a) - V(b)$$

Use the fundamental theorem of gradient : $\int_a^b \nabla V \cdot d\mathbf{l} = V(b) - V(a)$

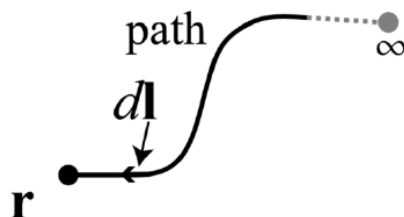
$$\int_a^b \nabla V \cdot d\mathbf{l} = - \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

$$\mathbf{E} = -\nabla V$$

Electric Potential:

- (1) Electric potential is different from electric potential energy. Unit of electric potential is Newton-meter per Coulomb ($\frac{\text{N}\cdot\text{m}}{\text{C}}$) or Volt.
- (2) The potential obeys superposition principle, i.e., the potential due to several charges is equal to the sum of the potentials due to individual ones: $V = V_1 + V_2 + \dots$
- (3) If one knows the electrical potential (a scalar quantity), the electric field (a vector quantity) can be calculated
- (4) The electric field is a vector quantity, but we still get all the information from the potential (a scalar quantity). This is because different components are interrelated: $\nabla \times \mathbf{E} = 0$, i.e., $\frac{\partial E_x}{\partial y} = \frac{\partial E_y}{\partial x}$; $\frac{\partial E_z}{\partial y} = \frac{\partial E_y}{\partial z}$; $\frac{\partial E_x}{\partial z} = \frac{\partial E_z}{\partial x}$;
- (5) $V(\mathbf{b}) - V(\mathbf{a}) = -\int_a^b \mathbf{E} \cdot d\mathbf{l}$. Absolute potential cannot be defined. In electrostatics, usually one takes the reference point to infinity and set the potential at infinity to zero, that is, take $V(\mathbf{a}) = V(\infty) = 0$. Also if $V(\mathbf{b}) = V(\mathbf{r})$,

$$V(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}$$



Electric Potential due to a point charge at origin:

Electric field $\mathbf{E}(\mathbf{r}_1)$ at $\mathbf{r}_1$ due to a single point charge q at origin:

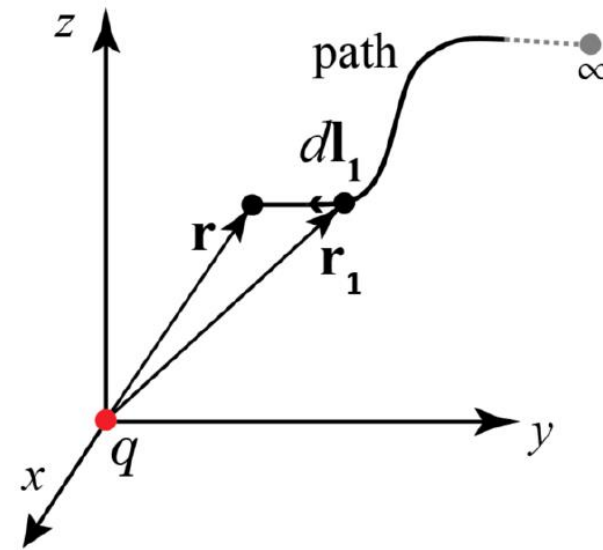
$$\mathbf{E}(\mathbf{r}_1) = \frac{1}{4\pi\epsilon_0} \frac{q}{r_1^2} \hat{\mathbf{r}}_1$$

Electric potential $V(\mathbf{r})$ at $\mathbf{r}$ due to a single point charge q at origin:

$$V(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l} = - \int_{\infty}^{\mathbf{r}} \mathbf{E}(\mathbf{r}_1) \cdot d\mathbf{l}_1$$

The line element is: $d\mathbf{l}_1 = dr_1 \hat{\mathbf{r}}_1 + r_1 d\theta_1 \hat{\boldsymbol{\theta}}_1 + r_1 \sin\theta_1 d\phi_1 \hat{\boldsymbol{\phi}}_1$

$$V(\mathbf{r}) = - \int_{\infty}^{\mathbf{r}} \frac{1}{4\pi\epsilon_0} \frac{q}{r_1^2} dr_1 = - \frac{q}{4\pi\epsilon_0} \int_{\infty}^{\mathbf{r}} \frac{1}{r_1^2} dr_1 = \frac{q}{4\pi\epsilon_0} \frac{1}{r}$$



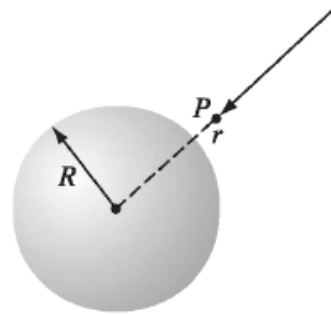
$$V(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \frac{1}{r}$$

Example : Find the potential inside and outside a spherical shell of radius R that carries a uniform surface charge. Set the reference point at infinity.

Solution

From Gauss's law, the field outside is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}},$$



where q is the total charge on the sphere. The field inside is zero.

For points outside the sphere ($r > R$),

$$V(r) = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l} = \frac{-1}{4\pi\epsilon_0} \int_{\infty}^r \frac{q}{r'^2} dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r'} \Big|_{\infty}^r = \frac{1}{4\pi\epsilon_0} \frac{q}{r}.$$

To calculate the potential inside the sphere ($r < R$), we must break the integral into two pieces, using in each region the field

$$V(r) = \frac{-1}{4\pi\epsilon_0} \int_{\infty}^R \frac{q}{r'^2} dr' - \int_R^r (0) dr' = \frac{1}{4\pi\epsilon_0} \frac{q}{r'} \Big|_{\infty}^R + 0 = \frac{1}{4\pi\epsilon_0} \frac{q}{R}.$$

Problem : Find the potential inside and outside a uniformly charged solid sphere whose radius is R and whose total charge is q . Use infinity as your reference point.

Solution : $V(r) = - \int_{\infty}^r \mathbf{E} \cdot d\mathbf{l}.$

Outside the sphere ($r > R$) : $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}.$

So for $r > R$: $V(r) = - \int_{\infty}^r \left(\frac{1}{4\pi\epsilon_0} \frac{q}{\bar{r}^2} \right) d\bar{r} = \frac{1}{4\pi\epsilon_0} q \left(\frac{1}{\bar{r}} \right) \Big|_{\infty}^r = \boxed{\frac{q}{4\pi\epsilon_0} \frac{1}{r}},$

Inside the sphere ($r < R$) : $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} r \hat{\mathbf{r}}.$

for $r < R$: $V(r) = - \int_{\infty}^R \left(\frac{1}{4\pi\epsilon_0} \frac{q}{\bar{r}^2} \right) d\bar{r} - \int_R^r \left(\frac{1}{4\pi\epsilon_0} \frac{q}{R^3} \bar{r} \right) d\bar{r}$
 $= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{R} - \frac{1}{R^3} \left(\frac{r^2 - R^2}{2} \right) \right]$
 $= \boxed{\frac{q}{4\pi\epsilon_0} \frac{1}{2R} \left(3 - \frac{r^2}{R^2} \right)}.$

Electric Potential due to localized charge distribution:

Potential due to a point charge q at origin:

$$V(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \frac{1}{r}$$

Potential due to a point charge q at $\mathbf{r}'$:

$$V(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \frac{1}{z}$$

Potential due to a collection of point charges

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{z_i}$$

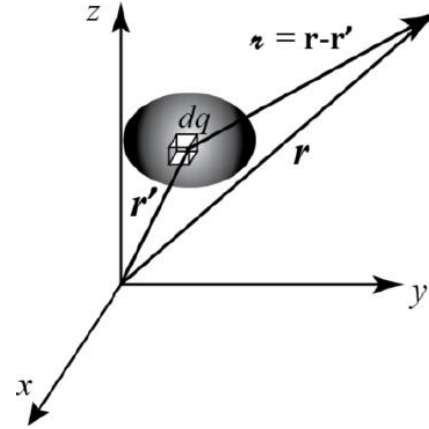
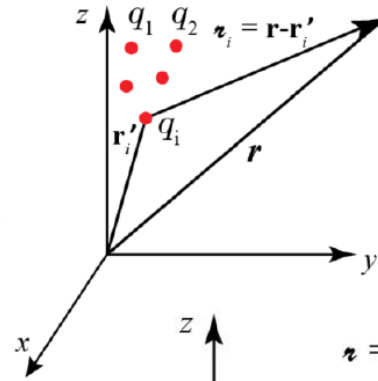
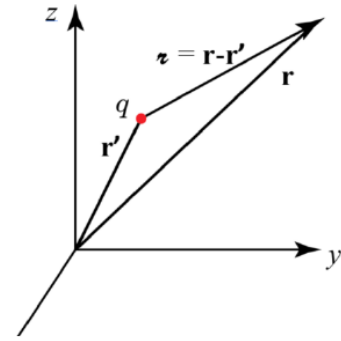
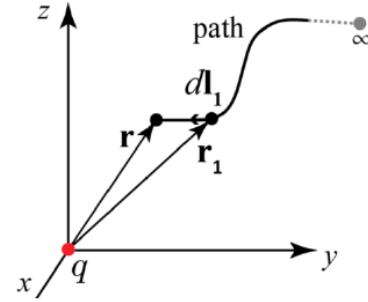
Potential due to a continuous charge distribution is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{z}$$

For a line charge $dq = \lambda(\mathbf{r}')dl'$

For a surface charge $dq = \sigma(\mathbf{r}')da'$

For a volume charge $dq = \rho(\mathbf{r}')d\tau'$

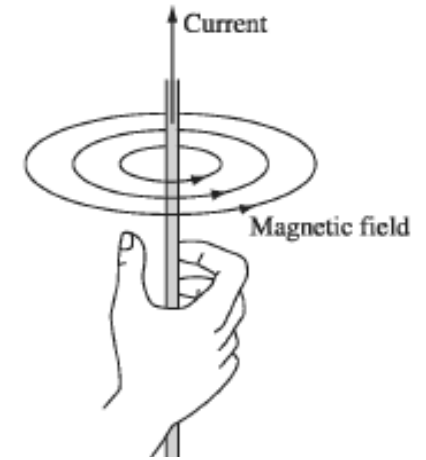


Forces on a Current-Carrying Wire

MIT Physics Lecture
Demonstration Group

Current produces magnetic field

Magnetic Field interacts with moving charges



Magnetostatics

Stationary charges $\longrightarrow$ Constant Electric field $\longrightarrow$ Electrostatics

Steady currents $\longrightarrow$ Constant Magnetic field $\longrightarrow$ Magnetostatics

Non-stationary charges
and/or Non-steady currents $\longrightarrow$ Changing Electric and
Magnetic field $\longrightarrow$ Electrodynamics

Force on a point charge Q :

Electric Force: $\mathbf{F}_{\text{elec}} = Q\mathbf{E}$

Magnetic Force: $\mathbf{F}_{\text{mag}} = Q(\mathbf{v} \times \mathbf{B})$ **Lorentz Force Law**

Total Force: $\mathbf{F} = \mathbf{F}_{\text{elec}} + \mathbf{F}_{\text{mag}} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$

Work done by magnetic forces

$$W_{\text{mag}} = \int \mathbf{F}_{\text{mag}} \cdot d\mathbf{l} = \int Q(\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l} = \int Q(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt = 0$$

- Magnetic forces do no work
- Magnetic forces can change the direction in which a particle moves.
- Magnetic forces do not change the speed with which a particle moves.

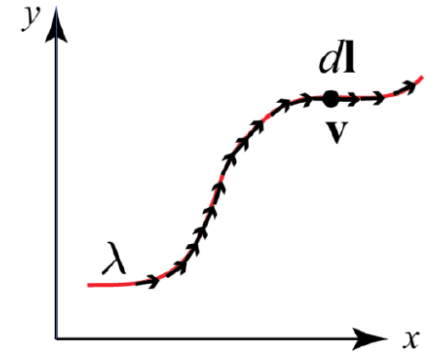
How to calculate $\mathbf{F}_{\text{mag}}$ and the magnetic field $\mathbf{B}$??

Current: Measure of charge flow

- Current is charge flow per unit time $I = \frac{dq}{dt}$
- It is measured in Coulombs per second, or Amperes (A).
- Current is a vector, the direction of current is in the direction of charge-flow
- Conventionally, this is the direction opposite to the flow of electrons.

Charge flowing in a wire :

If uniform line charge density λ passing through a wire with a velocity $\mathbf{v}$, then at time dt , $d\mathbf{l} = \mathbf{v}dt$ line charge passes through any point on the wire.



The net amount of charge passes through any point at dt time is $dq = \lambda dl$

$$\text{Therefore, } \mathbf{I} = \frac{dq}{dt} \hat{n} = \lambda \frac{d\mathbf{l}}{dt} = \lambda \mathbf{v}$$

If Magnetic field $\mathbf{B}$ present,

The magnetic force on a segment of current-carrying wire is

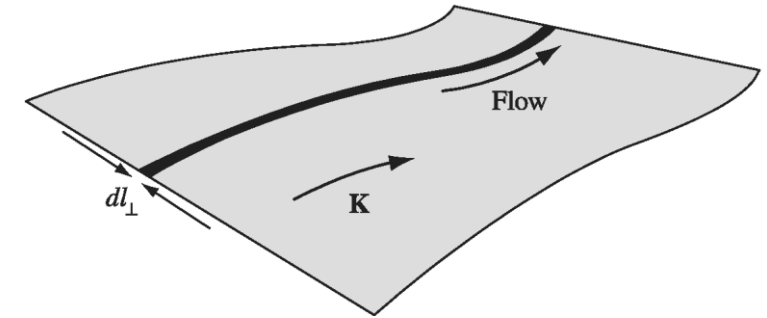
$$\mathbf{F}_{mag} = \int (\mathbf{v} \times \mathbf{B}) \lambda dl = \int (\lambda \mathbf{v} \times \mathbf{B}) dl = \int (\mathbf{I} \times \mathbf{B}) dl$$

Charge flowing on a surface:

When charge flows over a *surface*, we describe it by the surface current density $\mathbf{K}$, defined as follows:

Consider a "ribbon" of infinitesimal width $dl_{\perp}$, running parallel to the flow. If the current in this ribbon is $d\mathbf{I}$, the surface current density is $\mathbf{K} = \frac{d\mathbf{I}}{dl_{\perp}}$

This is current per unit width perpendicular to the flow.



If uniform surface charge density σ passing through on the surface with a velocity $\mathbf{v}$, then at time dt , $dq = \sigma[(\mathbf{v}dt)dl_{\perp}]$ charge passes through the ribbon of width $dl_{\perp}$.

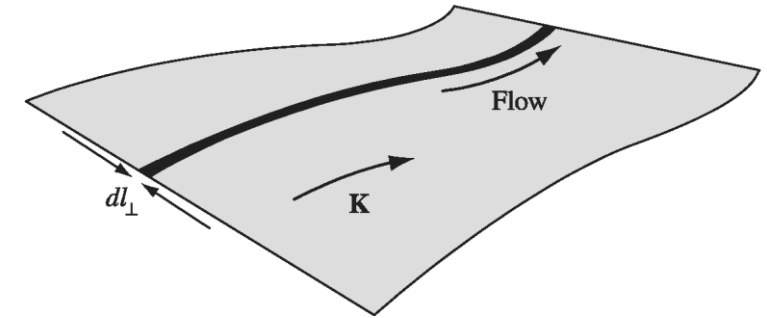
The current passes through ribbon is $d\mathbf{I} = \frac{dq}{dt} \hat{n} = \sigma \mathbf{v} dl_{\perp}$

$$\mathbf{K} = \frac{d\mathbf{I}}{dl_{\perp}} = \sigma \mathbf{v}$$

Charge flowing on a surface:

In presence of magnetic field $\mathbf{B}$,
The magnetic force on the surface current is

$$\begin{aligned}\mathbf{F}_{mag} &= \int (\mathbf{v} \times \mathbf{B}) dq = \int (\mathbf{v} \times \mathbf{B}) \sigma da \\ &= \int (\sigma \mathbf{v} \times \mathbf{B}) da = \int (\mathbf{K} \times \mathbf{B}) da\end{aligned}$$

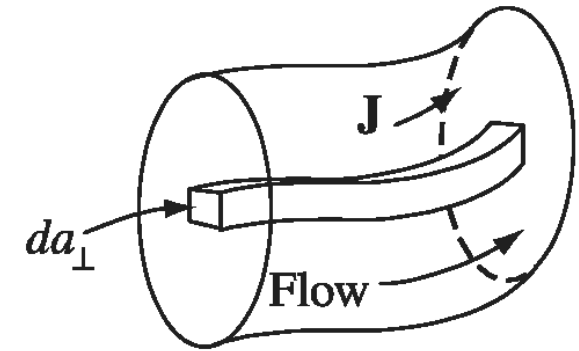


Charge flowing through volume:

When charge flows through *volume*, we describe it by the volume current density $\mathbf{J}$, defined as follows:

Consider a “tube” of infinitesimal cross section $da_{\perp}$, running parallel to the flow. If the current in this tube is $d\mathbf{I}$, the volume current density is $\mathbf{J} = \frac{d\mathbf{I}}{da_{\perp}}$

Which is current per unit area perpendicular to the flow.



If uniform volume charge density ρ passing through the volume with a velocity $\mathbf{v}$, then at time dt , $dq = \rho(\mathbf{v}dt) da_{\perp}$ charge passes through the tube of cross section $da_{\perp}$.

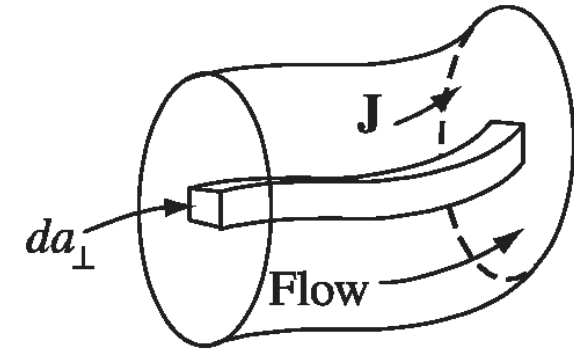
The current passes through tube is $d\mathbf{I} = \frac{dq}{dt} \hat{n} = \rho \mathbf{v} da_{\perp}$

$$\mathbf{J} = \frac{d\mathbf{I}}{da_{\perp}} = \rho \mathbf{v}$$

Charge flowing through volume:

In presence of magnetic field $\mathbf{B}$,
The magnetic force on the volume current is

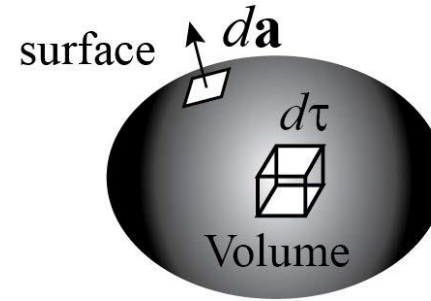
$$\begin{aligned}\mathbf{F}_{mag} &= \int (\mathbf{v} \times \mathbf{B}) dq = \int (\mathbf{v} \times \mathbf{B}) \rho d\tau \\ &= \int (\rho \mathbf{v} \times \mathbf{B}) d\tau = \int (\mathbf{J} \times \mathbf{B}) d\tau\end{aligned}$$



The Continuity Equation (Conservation of Charge):

Consider a closed surface and current is flowing out

Current = Volume current density crossing a surface
= Total charge crossing a surface per unit time



$$\mathbf{J} = \frac{d\mathbf{I}}{da_{\perp}} \quad \Rightarrow \quad I = \int_S \mathbf{J} \, da_{\perp} \quad \Rightarrow \quad I = \int_S \mathbf{J} \cdot d\mathbf{a}$$

$$I = \oint_S \mathbf{J} \cdot d\mathbf{a} \quad \Rightarrow \quad \begin{array}{l} \text{Total charge per unit time crossing a closed surface} \\ \text{Or, Total charge leaving the volume } V \text{ per unit time} \end{array}$$

$$\oint_S \mathbf{J} \cdot d\mathbf{a} = \int_V (\nabla \cdot \mathbf{J}) d\tau \quad \text{But, total charge per unit time leaving the volume } V \text{ is } -\frac{d}{dt} \left(\int_V \rho d\tau \right) = -\int_V \frac{d\rho}{dt} d\tau$$

$$\int_V (\nabla \cdot \mathbf{J}) d\tau = -\int_V \frac{d\rho}{dt} d\tau \quad \Rightarrow \quad \boxed{\nabla \cdot \mathbf{J} = -\frac{d\rho}{dt}} \quad \text{The Continuity Equation}$$

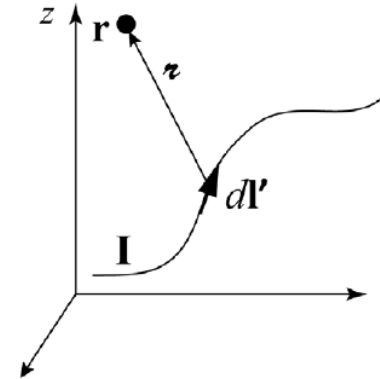
Magnetic field :

Steady currents produce magnetic fields that are constant in time; the theory of steady currents is called **magnetostatics**

$$\text{Steady currents} \Rightarrow \frac{\partial \mathbf{J}}{\partial t} = 0$$

$$\text{The Biot-Savart law: } \mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{I} \times \hat{\mathbf{r}}}{r^2} dl = \frac{\mu_0}{4\pi} I \int \frac{d\mathbf{l}' \times \hat{\mathbf{r}}}{r^2}$$

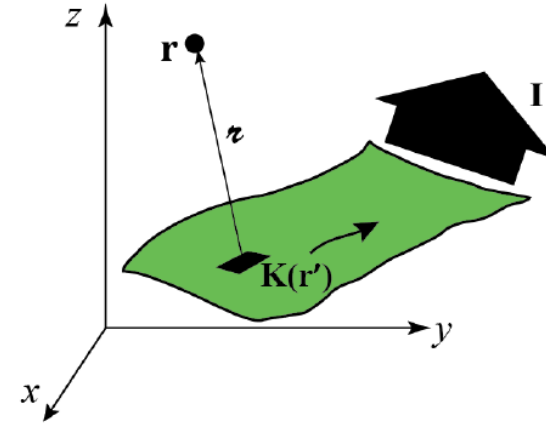
- μ_0 is the permeability of free space
- $\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$
- The unit of magnetic field is Newton per Ampere-meter, or Tesla
- 1 Tesla is a very strong magnetic field. Earth's magnetic field is about 10^{-4} times smaller
- Biot-Savart law for magnetic field is analogous to Coulomb's law for electric field



The Biot-Savart Law:

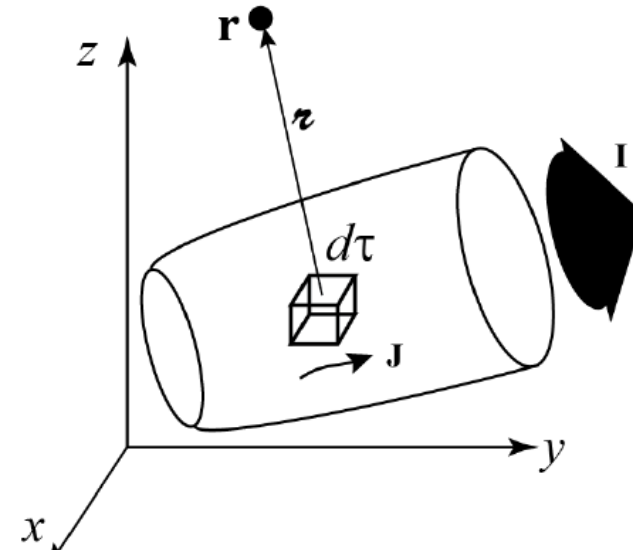
The magnetic field produced by a surface current:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} da'$$



The magnetic field produced by a volume current:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau'$$



The Biot-Savart Law:

Ex. 1: Calculate the magnetic field due to a long straight wire carrying a steady current I .

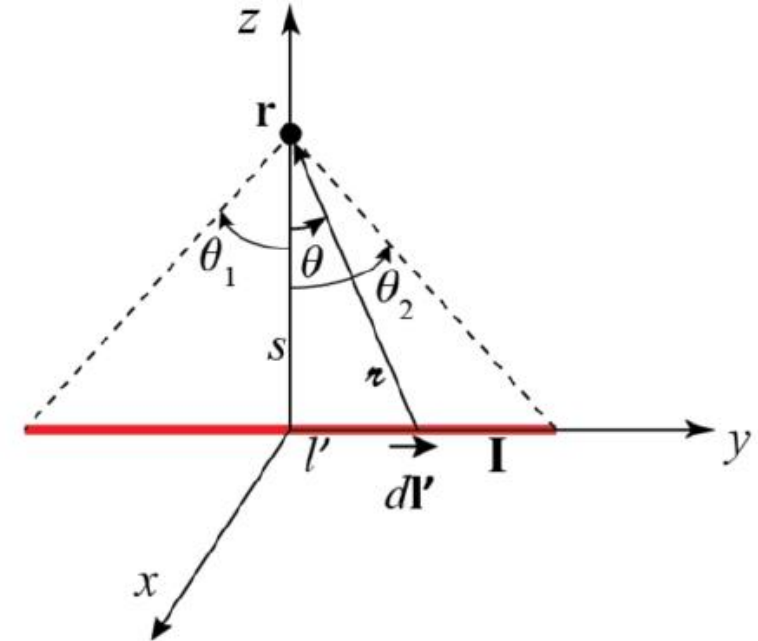
$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} I \int \frac{d\mathbf{l}' \times \hat{\mathbf{r}}}{r^2} \quad \text{Clearly} \quad \mathbf{B}(\mathbf{r}) = B \hat{\mathbf{x}}$$

$$|d\mathbf{l}' \times \hat{\mathbf{r}}| = dl' \cos\theta$$

$$l' = s \tan\theta \quad \Rightarrow \quad dl' = \frac{s}{\cos^2\theta} d\theta$$

$$s = r \cos\theta \quad \Rightarrow \quad \frac{1}{r^2} = \frac{\cos^2\theta}{s^2}$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} I \int_{\theta_1}^{\theta_2} \left(\frac{\cos^2\theta}{s^2} \right) \left(\frac{s}{\cos^2\theta} \right) \cos\theta d\theta \hat{\mathbf{x}} = \frac{\mu_0 I}{4\pi s} \int_{-\theta_1}^{\theta_2} \cos\theta d\theta \hat{\mathbf{x}} = \frac{\mu_0 I}{4\pi s} (\sin\theta_2 + \sin\theta_1) \hat{\mathbf{x}}$$



The Biot-Savart Law:

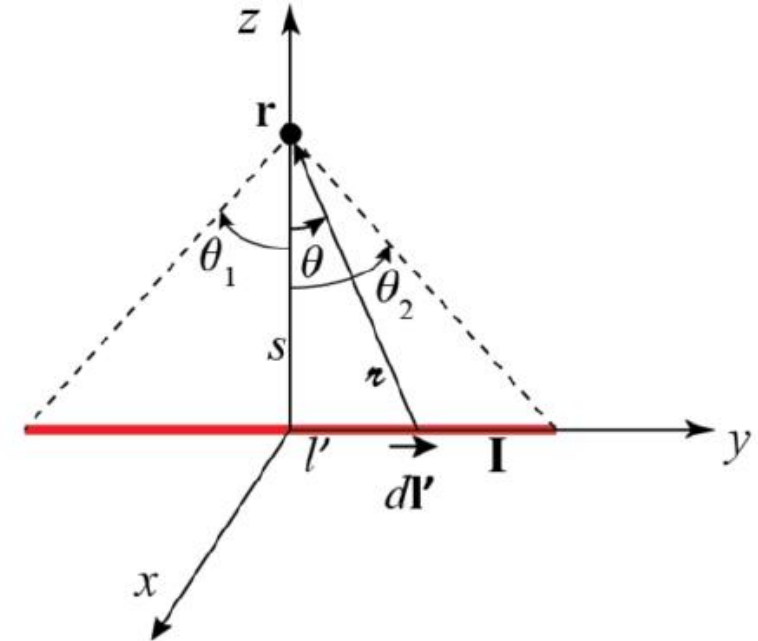
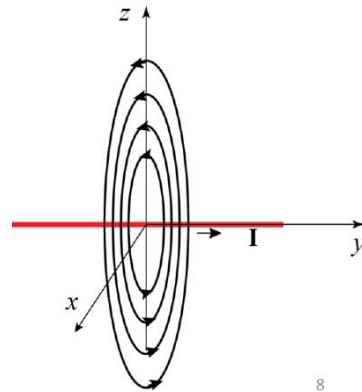
Ex. 1: Calculate the magnetic field due to a long straight wire carrying a steady current I .

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi s} (\sin\theta_2 + \sin\theta_1) \hat{\mathbf{x}}$$

What will be the field due to an infinite wire?

$$\theta_1 = \frac{\pi}{2} \quad \theta_2 = \frac{\pi}{2}$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi s} (\sin\theta_2 + \sin\theta_1) \hat{\mathbf{x}} = \frac{\mu_0 I}{4\pi s} (1 + 1) \hat{\mathbf{x}} = \frac{\mu_0 I}{2\pi s} \hat{\mathbf{x}}$$



The Biot-Savart Law:

Ex.2 : Find the magnetic field $\mathbf{B}(\mathbf{r})$ field above the center of a current loop of radius R , which carries a steady current I .

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} I \int \frac{d\mathbf{l}' \times \hat{\mathbf{r}}}{r^2}$$

The line element $d\mathbf{l}'$ produces the field $d\mathbf{B}$ at $\mathbf{r}$.

Note that the horizontal components of this field calculated from different parts of the loop cancels each other.(no need to bother)

The vertical component of this field is:
$$dB(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \frac{dl'}{r^2} \cos\theta$$

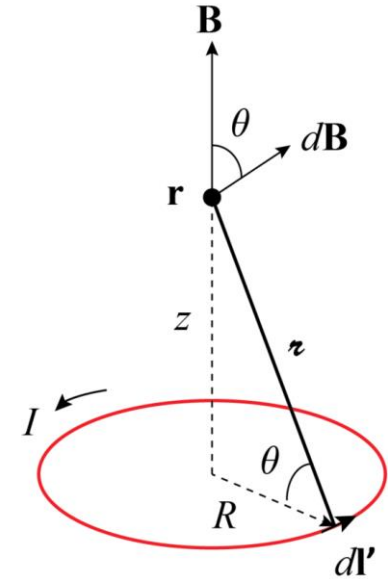
The total field, which is in the z direction, is

$$\mathbf{B}(z) = \frac{\mu_0 I}{4\pi} \int \frac{\cos\theta}{r^2} dl' = \frac{\mu_0 I \cos\theta}{4\pi r^2} \int dl' = \frac{\mu_0 I R}{4\pi r^3} (2\pi R) = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}}$$

What is the field at the center?

$$\mathbf{B}(0) = \frac{\mu_0 I R^2}{2 R^3} = \frac{\mu_0 I}{2R}$$

Useful to obtain the magnetic field produced by a solenoid



The Divergence and Curl of $\mathbf{B}$:

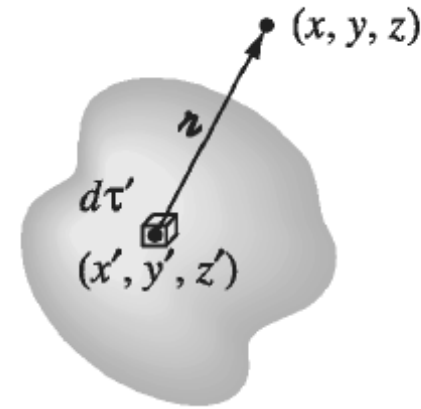
$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau'$$

$\mathbf{B}$ is a function of (x, y, z) ,

$\mathbf{J}$ is a function of (x', y', z') ,

$$\mathbf{r} = (x - x') \hat{\mathbf{x}} + (y - y') \hat{\mathbf{y}} + (z - z') \hat{\mathbf{z}},$$

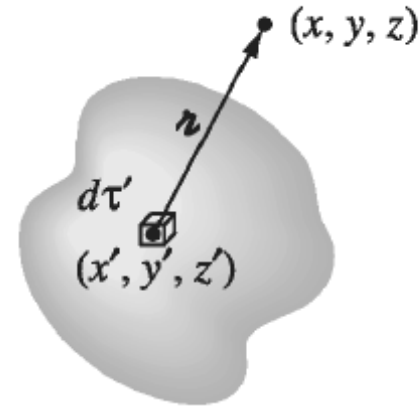
$$d\tau' = dx' dy' dz'.$$



The integration is over the *primed* coordinates; the divergence and the curl of $\mathbf{B}$ are with respect to the *unprimed* coordinates.

The Divergence of $\mathbf{B}$:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau'$$



$$\nabla \cdot \mathbf{B} = \frac{\mu_0}{4\pi} \int \nabla \cdot \left(\mathbf{J} \times \frac{\hat{\mathbf{r}}}{r^2} \right) d\tau'$$

$$\nabla \cdot \left(\mathbf{J} \times \frac{\hat{\mathbf{r}}}{r^2} \right) = \frac{\hat{\mathbf{r}}}{r^2} \cdot (\nabla \times \mathbf{J}) - \mathbf{J} \cdot \left(\nabla \times \frac{\hat{\mathbf{r}}}{r^2} \right)$$

$\nabla \times \mathbf{J} = 0$, because $\mathbf{J}$ doesn't depend on the unprimed variables

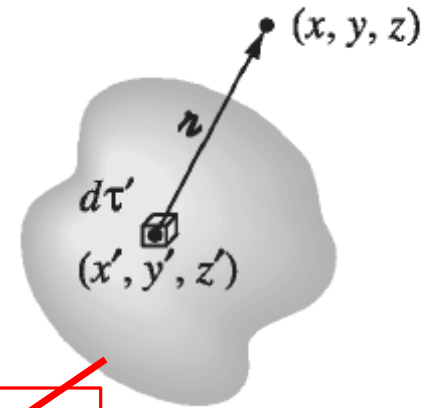
$$\nabla \times (\hat{\mathbf{r}}/r^2) = \mathbf{0}$$

$$\nabla \cdot \mathbf{B} = 0.$$

The Curl of $\mathbf{B}$:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{r}}}{r^2} d\tau'$$

$$\nabla \times \mathbf{B} = \frac{\mu_0}{4\pi} \int \nabla \times \left(\mathbf{J} \times \frac{\hat{\mathbf{r}}}{r^2} \right) d\tau'$$



$$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})$$

$$\nabla \times \left(\mathbf{J} \times \frac{\hat{\mathbf{r}}}{r^2} \right) = \mathbf{J} \left(\nabla \cdot \frac{\hat{\mathbf{r}}}{r^2} \right) - (\mathbf{J} \cdot \nabla) \frac{\hat{\mathbf{r}}}{r^2}$$

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi \delta^3(\mathbf{r})$$

$$\int (\text{Second term}) d\tau = 0$$

$$\nabla \times \mathbf{B} = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}') 4\pi \delta^3(\mathbf{r} - \mathbf{r}') d\tau' = \mu_0 \mathbf{J}(\mathbf{r})$$

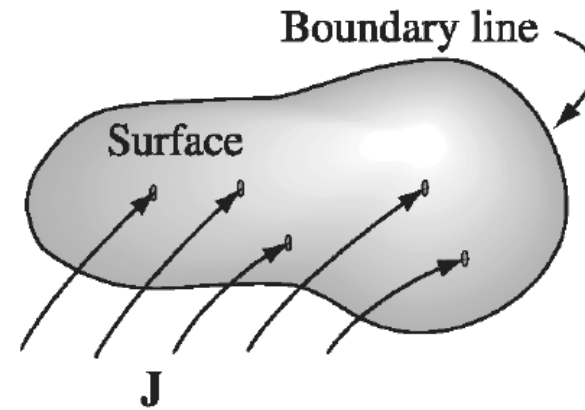
$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J},$$

Ampere's Law

Ampere's Law:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

$$\int (\nabla \times \mathbf{B}) \cdot d\mathbf{a} = \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \int \mathbf{J} \cdot d\mathbf{a}.$$



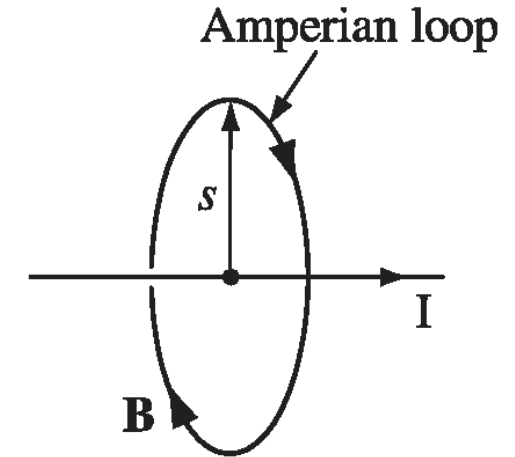
$\int \mathbf{J} \cdot d\mathbf{a} \Rightarrow$ the total current passing through the surface, which we call I_{enc} (the current enclosed by the Amperian loop).

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}.$$

Ampere's Law:

Example : Find the magnetic field a distance s from a long straight wire, carrying a steady current I

We know the direction of $\mathbf{B}$ is "circumferential," circling around the wire as indicated by the right-hand rule. By symmetry, the magnitude of $\mathbf{B}$ is constant around an Amperian loop of radius s , centered on the wire.



Applying Ampere's law we get

$$\oint \mathbf{B} \cdot d\mathbf{l} = B \oint dl = B2\pi s = \mu_0 I_{\text{enc}} = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi s}.$$

Ampere's Law:

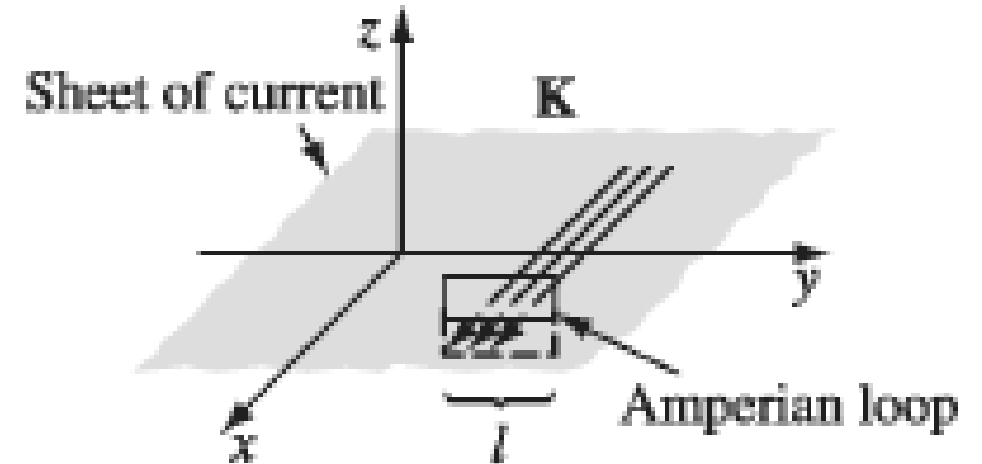
Example 2. Find the magnetic field due to an infinite uniform surface current $\mathbf{K} = K \hat{x}$, flowing over the xy -plane.

Answer :

What the direction of the field ?

$$\oint \mathbf{B} \cdot d\mathbf{l} = 2Bl = \mu_0 I_{\text{enc}} = \mu_0 K l,$$

$$\mathbf{B} = \begin{cases} +(\mu_0/2)K \hat{y} & \text{for } z < 0, \\ -(\mu_0/2)K \hat{y} & \text{for } z > 0. \end{cases}$$



Ampere's Law:

Example 3. Find the magnetic field of a very long solenoid, consisting of n closely wound turns per unit length on a cylinder of radius R , each carrying a steady current I

Answer :

What the direction of the field ?

1. radial component = 0

2. circumferential" component? $\oint \mathbf{B} \cdot d\mathbf{l} = B_\phi(2\pi s) = \mu_0 I_{\text{enc}} = 0,$

3 Field is along the axis direction !!!

Outside the solenoid

$$\oint \mathbf{B} \cdot d\mathbf{l} = [B(a) - B(b)]L = \mu_0 I_{\text{enc}} = 0,$$
$$B(a) = B(b).$$

Inside the solenoid

$$\oint \mathbf{B} \cdot d\mathbf{l} = BL = \mu_0 I_{\text{enc}} = \mu_0 n I L,$$

$$\mathbf{B} = \begin{cases} \mu_0 n I \hat{\mathbf{z}}, & \text{inside the solenoid,} \\ \mathbf{0}, & \text{outside the solenoid.} \end{cases}$$

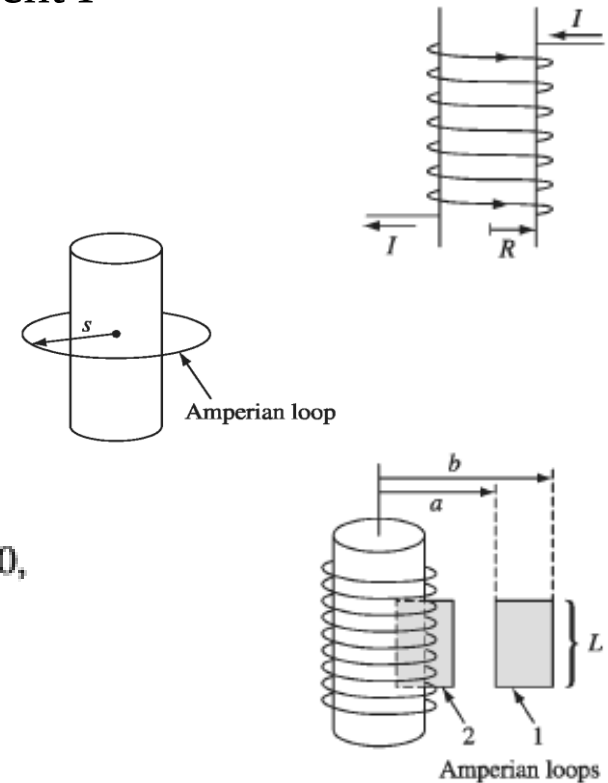
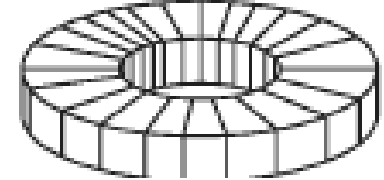


FIGURE 5.37

Ampere's Law:

Example 4: A toroidal coil consists of a circular ring, or "donut," around which a long wire is wrapped. The winding is uniform and tight enough so that each turn can be considered a plane closed loop. Determine the magnetic field produced by this coil when I current is passing through it.



Lets take point r in the xz -plane, so its Cartesian components are $(x, 0, z)$ and apply Biot-Savart Law

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{\mathbf{I} \times \hat{\mathbf{r}}}{r^3} dl'$$

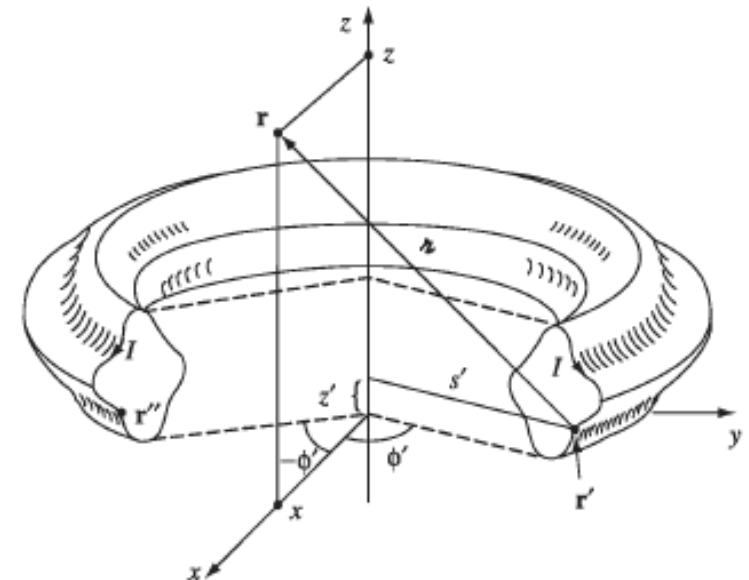
The source coordinates are

$$\mathbf{r}' = (s' \cos \phi', s' \sin \phi', z')$$

$$\mathbf{r} = (x - s' \cos \phi', -s' \sin \phi', z - z').$$

Since the current has no ϕ component, $\mathbf{I} = I_s \hat{\mathbf{s}} + I_z \hat{\mathbf{z}}$,

$$\mathbf{I} = (I_s \cos \phi', I_s \sin \phi', I_z).$$



$$\mathbf{I} \times \hat{\mathbf{r}} = \begin{bmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ I_s \cos \phi' & I_s \sin \phi' & I_z \\ (x - s' \cos \phi') & (-s' \sin \phi') & (z - z') \end{bmatrix} = \left[\sin \phi' (I_s (z - z') + s' I_z) \right] \hat{\mathbf{x}} + \left[I_z (x - s' \cos \phi') - I_s \cos \phi' (z - z') \right] \hat{\mathbf{y}} + \left[-I_s x \sin \phi' \right] \hat{\mathbf{z}}.$$

A similar dl component will be there at $-\phi'$

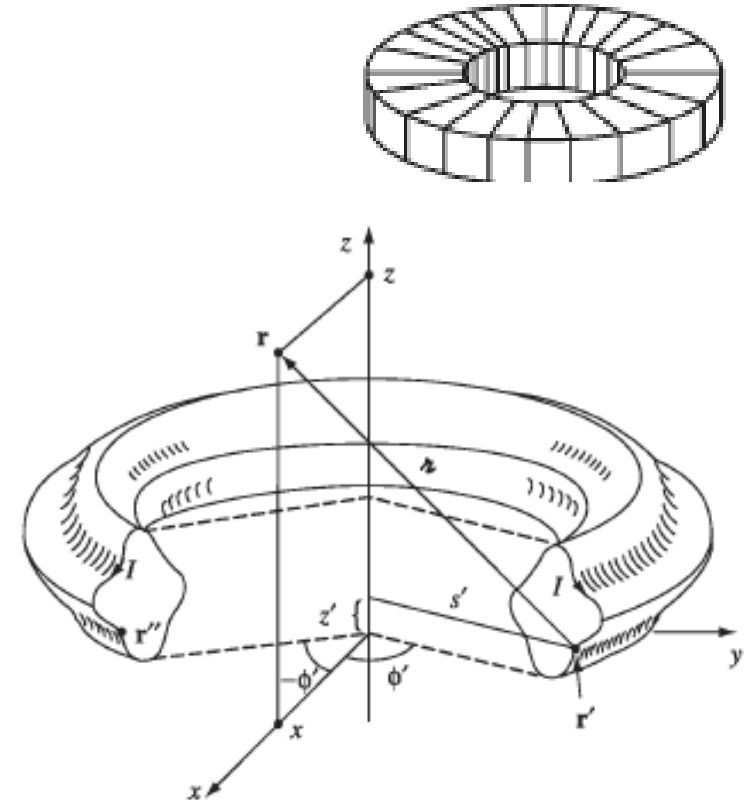
Ampere's Law:

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$$B2\pi s = \mu_0 I_{\text{enc}},$$

$$\mathbf{B}(\mathbf{r}) = \begin{cases} \frac{\mu_0 N I}{2\pi s} \hat{\phi}, & \text{for points inside the coil,} \\ 0, & \text{for points outside the coil,} \end{cases}$$

where N is the total number of turns.



Electrodynamics:

Force is need to flow the charge

The current density $\mathbf{J}$ is proportional to the *force per unit charge* $\mathbf{f}$:

$$\mathbf{J} \propto \mathbf{f}$$

or $\mathbf{J} = \sigma \mathbf{f}$ σ is called conductivity of the system

the *force per unit charge* $\mathbf{f} = (\mathbf{E} + \mathbf{v} \times \mathbf{B})$

$$\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

If v is sufficiently small $\mathbf{J} = \sigma(\mathbf{E})$ **(Ohm's Law)**

$\rho = 1/\sigma$ is called the resistivity

Example. Two long coaxial metal cylinders (radii a and b) are separated by material of conductivity σ . If they are maintained at a potential difference V , what current flows from one to the other, in a length L ?

Solution

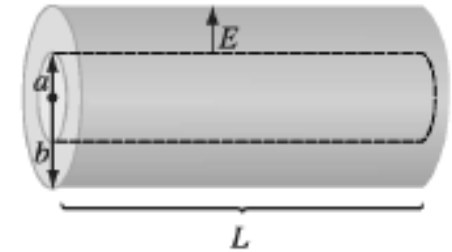
The field between the cylinders is

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{\mathbf{s}}, \quad \text{where } \lambda \text{ is the charge per unit length on the inner cylinder.}$$

The current
$$I = \int \mathbf{J} \cdot d\mathbf{a} = \sigma \int \mathbf{E} \cdot d\mathbf{a} = \frac{\sigma}{\epsilon_0} \lambda L.$$

$$V = - \int_b^a \mathbf{E} \cdot d\mathbf{l} = \frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{b}{a} \right)$$

$$I = \frac{2\pi\sigma L}{\ln(b/a)} V.$$



Electrodynamics:

What makes the current passes through the circuit ?

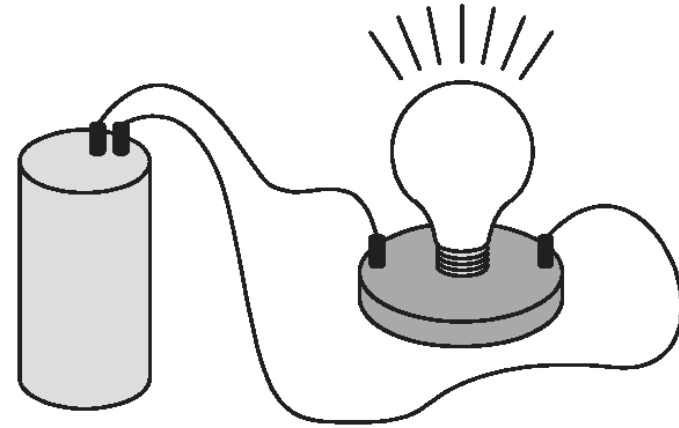
$$\mathbf{F} = \mathbf{f}_s + \mathbf{E}$$

Net effect of the force on the circuit per unit charge is

$$\mathcal{E} = \oint \mathbf{F} \cdot d\mathbf{l}$$

$$= \oint \mathbf{f}_s \cdot d\mathbf{l}$$

$\mathcal{E}$ is called electromotive force of *emf*



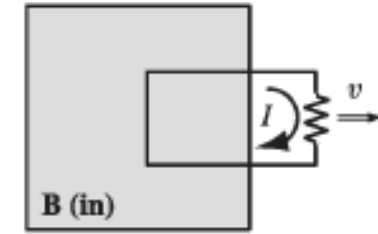
If there is no resistance inside the source $\Rightarrow$ no net force on the charge $\mathbf{F} = \mathbf{f}_s + \mathbf{E} = \mathbf{0}$

$$V = - \int_a^b \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{f}_s \cdot d\mathbf{l} = \mathcal{E}$$

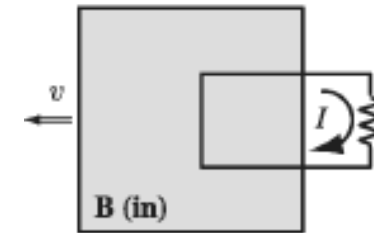
Faraday's Experiment

In 1831 Michael Faraday reported a series of experiments

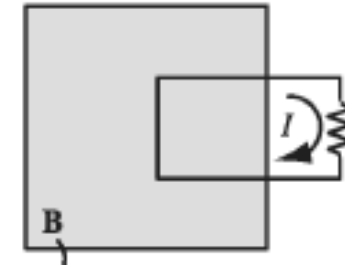
Experiment 1. He pulled a loop of wire to the right through a magnetic field. A current flowed in the loop



Experiment 2. He moved the *magnet* to the *left*, holding the loop still. Again, a current flowed in the loop.



Experiment 3. With both the loop and the magnet at rest, he changed the *strength* of the Magnetic field. Once again, current flowed in the loop.



Faraday's Experiment

Experiment 1. He pulled a loop of wire to the right through a magnetic field. A current flowed in the loop

The *emf* is generated due to the motion of the loop (motional *emf*)

The line segment *ab* feels the Lorentz force causing current in the clockwise direction

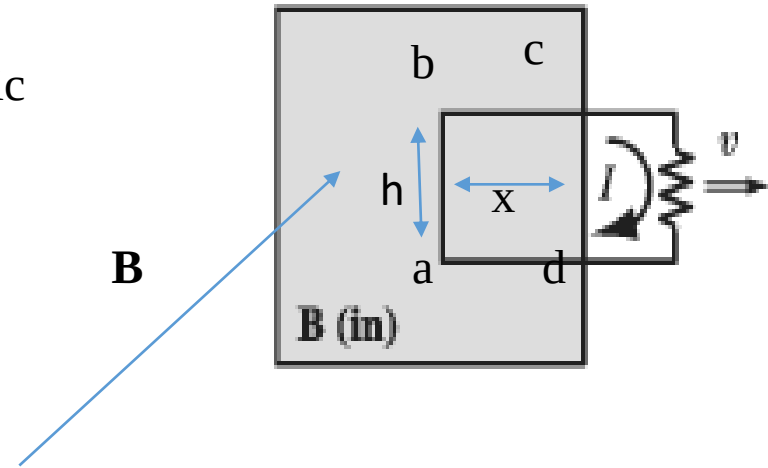
$$\mathbf{F} = \mathbf{v} \times \mathbf{B} = vB \sin 90^\circ = vB$$

$$\mathcal{E} = \oint \mathbf{F} \cdot d\mathbf{l} = vB h$$

The magnetic flux passes through the loop is $\phi = \int \mathbf{B} \cdot d\mathbf{a} = Bhx$

The change of flux due to pulling of the loop is $= \frac{d\phi}{dt} = Bh \frac{dx}{dt} = -Bhv$

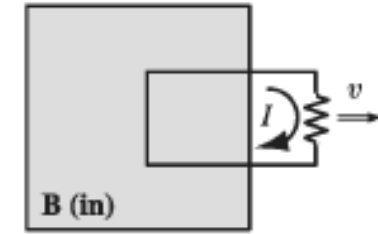
$$\mathcal{E} = -\frac{d\phi}{dt}$$



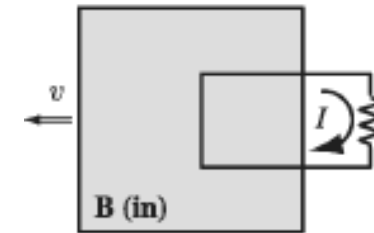
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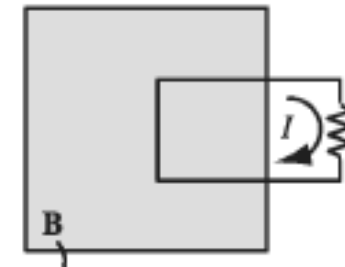
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Change in magnetic flux produces emf !!

$$\mathcal{E} = -\frac{d\phi}{dt}$$

Faraday's Experiment

$$\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\phi}{dt} = -\frac{d}{dt} \int \mathbf{B} \cdot d\mathbf{a}$$

$$\oint \mathbf{E} \cdot d\mathbf{l} = - \int \frac{d\mathbf{B}}{dt} \cdot d\mathbf{a}$$

$$\int (\nabla \times \mathbf{E}) \cdot d\mathbf{a} = - \int \frac{d\mathbf{B}}{dt} \cdot d\mathbf{a} \quad \nabla \times \mathbf{E} = -\frac{d\mathbf{B}}{dt} \quad \text{Faraday's Law in differential forms}$$

Faraday's law generalizes the electrostatic rule $\nabla \times \mathbf{E} = 0$ for static charge to time dependent regime for moving charges

The *divergence* of $\mathbf{E}$ is still given by Gauss's law $\nabla \cdot \mathbf{E} = 0$.

An induced electric field is generated due to the change of magnetic flux density and is determined by $-\frac{d\mathbf{B}}{dt}$

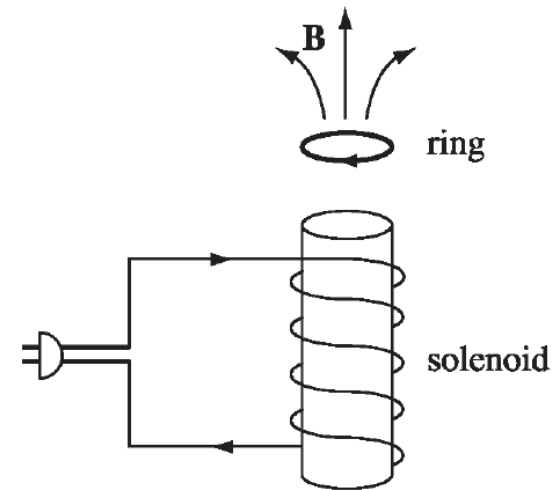
Faraday's Experiment

Whenever (and for whatever reason) the magnetic flux through a loop changes, an emf $\mathcal{E} = -\frac{d\Phi}{dt}$

What is the direction of current flow ??

Lenz's Law : The induced current will flow in such a direction that the flux *it* produces tends to cancel the change.

Simple Demonstration you may try at home :



How much electric field is produced due to the change of magnetic field ?

If there are no static charges

$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}.$$

The analog to Biot-Savart

$$\mathbf{E} = -\frac{1}{4\pi} \int \frac{(\partial \mathbf{B} / \partial t) \times \hat{\mathbf{r}}}{r^2} d\tau = -\frac{1}{4\pi} \frac{\partial}{\partial t} \int \frac{\mathbf{B} \times \hat{\mathbf{r}}}{r^2} d\tau,$$

$$\text{if symmetry permits, } \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}, \quad (\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}).$$

Maxwell's Equation

So far, we have encountered the following laws, specifying the divergence and curl of electric and magnetic fields:

$$(i) \quad \nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho \quad (\text{Gauss's law}),$$

$$(ii) \quad \nabla \cdot \mathbf{B} = 0 \quad (\text{no name}),$$

$$(iii) \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (\text{Faraday's law}),$$

$$(iv) \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad (\text{Ampère's law}).$$

If we carefully investigate the Faraday's law and Ampere's law, we can find some discrepancy

Take divergence in the both side of equation (iii)

$$\nabla \cdot (\nabla \times \mathbf{E}) = \nabla \cdot \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} (\nabla \cdot \mathbf{B}).$$

No issue

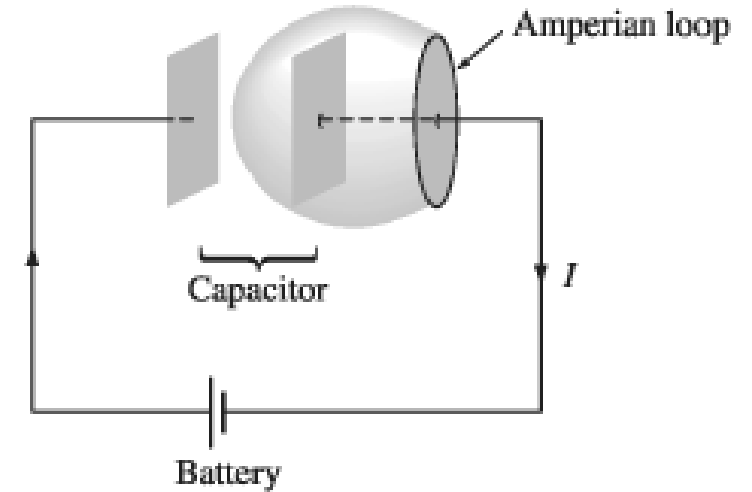
Take divergence in the both side of equation (iv)

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 (\nabla \cdot \mathbf{J})$$

Maxwell's Equation

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0(\nabla \cdot \mathbf{J})$$

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$



Essentially the right side of above equation *should be zero*, but *isn't*.

This can be fix using Gauss law and continuity equation as follows

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial t}(\epsilon_0 \nabla \cdot \mathbf{E}) = -\nabla \cdot \left(\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right).$$

Therefore if we combine $\epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$ with $\mathbf{J}$, then it will kill the extra divergence in the above equation

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

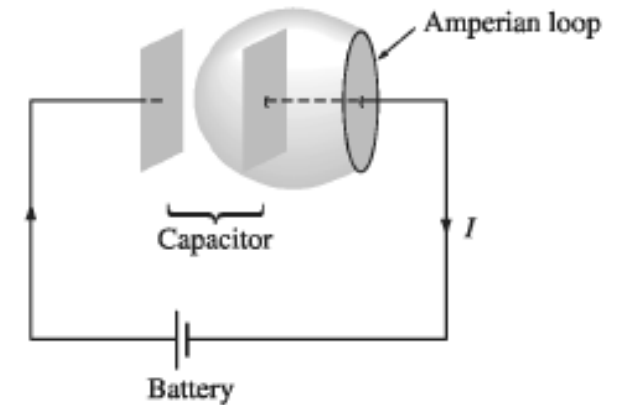
Displacement current: $\mathbf{J}_d \equiv \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$

If the capacitor plates are very close, then the electric field between them is

$$E = \frac{1}{\epsilon_0} \sigma = \frac{1}{\epsilon_0} \frac{Q}{A},$$

$$\frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 A} \frac{dQ}{dt} = \frac{1}{\epsilon_0 A} I.$$

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \int \left(\frac{\partial \mathbf{E}}{\partial t} \right) \cdot d\mathbf{a}.$$



If we choose the *flat* surface, then $E = 0$ and $I_{\text{enc}} = I$.

If we use the balloon-shaped surface, then

$$I_{\text{enc}} = 0, \text{ but } \int (\partial \mathbf{E} / \partial t) \cdot d\mathbf{a} = I / \epsilon_0.$$

Maxwell's Equation

Example: A uniform magnetic field $\mathbf{B}(t)$, pointing straight up, fills the shaded circular region of the figure. If $\mathbf{B}$ is changing with time, what is the induced electric field?

$\mathbf{E}$ points in the circumferential direction, just like the *magnetic* field inside a long straight wire carrying a uniform *current* density.

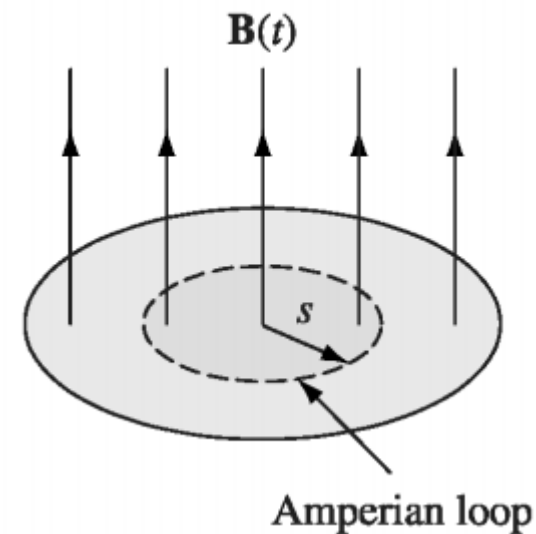
Faraday's law:

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}$$

Draw an Amperian loop of radius s ,

$$\left. \begin{array}{l} \oint \mathbf{E} \cdot d\mathbf{l} = E(2\pi s) \\ \Phi = \pi s^2 B(t) \end{array} \right\} \quad -\frac{d\Phi}{dt} = -\frac{d}{dt} (\pi s^2 B(t)) \quad \left. \begin{array}{l} E(2\pi s) = -\pi s^2 \frac{dB}{dt} \end{array} \right\} \Rightarrow \mathbf{E} = -\frac{s}{2} \frac{dB}{dt} \hat{\phi}.$$

If $\mathbf{B}$ is *increasing*, $\mathbf{E}$ runs *clockwise*, as viewed from above.



Maxwell's Equation

Example 2: A line charge λ is glued onto the rim of a wheel of radius b , which is then suspended horizontally, as shown in figure, so that it is free to rotate (the spokes are made of some nonconducting material-wood, maybe). In the central region, out to radius a , there is a uniform magnetic field B_0 pointing up. Now someone turns the field off. What happens?

The changing magnetic field will induce an electric field, curling around the axis of the wheel. This electric field exerts a force on the charges at the rim, and the wheel starts to turn. According to Lenz's law, it will rotate in such a direction that *its* field tends to restore the upward flux. The motion, then, is counterclockwise, as viewed from above.

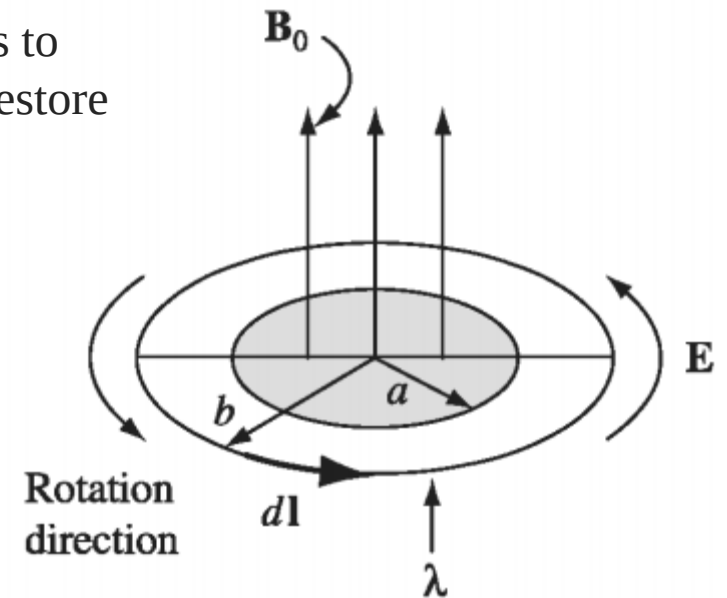
Faraday's law, applied to the loop at radius b , says

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt}$$
$$E(2\pi b) = -\pi a^2 \frac{dB}{dt} \quad \Rightarrow \quad \mathbf{E} = -\frac{a^2}{2b} \frac{dB}{dt} \hat{\phi}.$$

The torque on a segment of length dl is $(\mathbf{r} \times \mathbf{F})$, or $b \lambda E dl$.

The total torque on the wheel is therefore

$$N = b\lambda \left(-\frac{a^2}{2b} \frac{dB}{dt} \right) \oint dl = -b\lambda\pi a^2 \frac{dB}{dt}$$



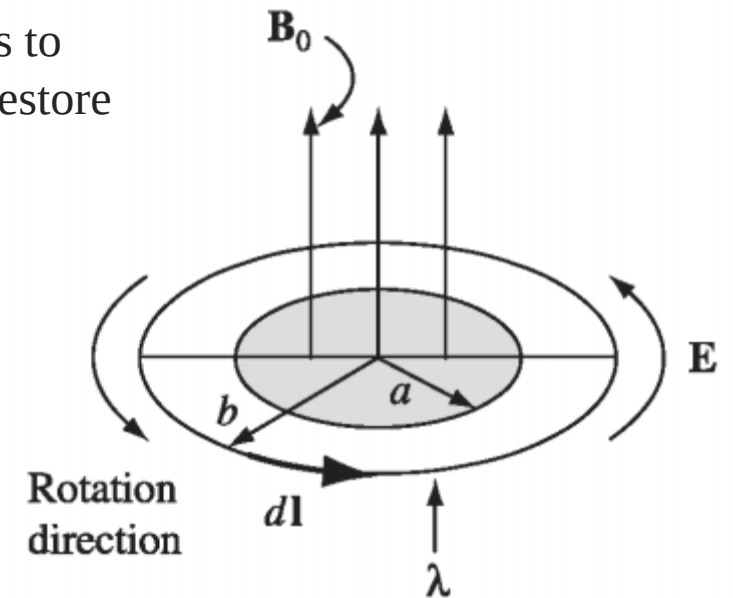
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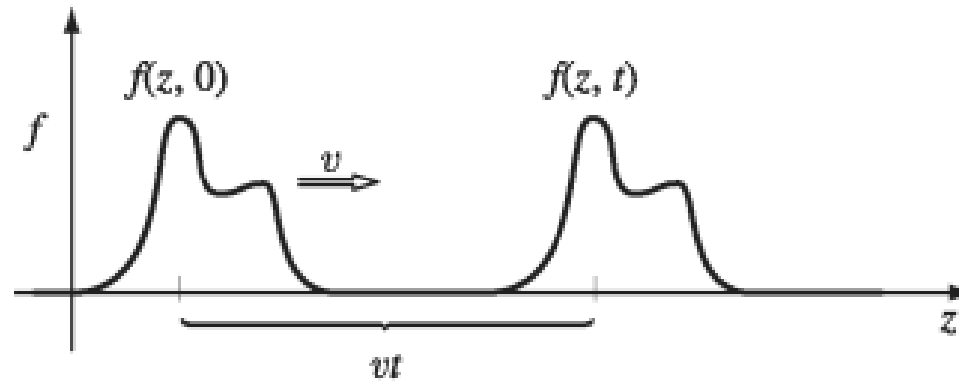
The angular momentum:

$$\int N dt = -\lambda \pi a^2 b \int_{B_0}^0 dB = \lambda \pi a^2 b B_0.$$



Electromagnetic Waves

What is a **WAVE** ?



How to represent a **WAVE** mathematically ?

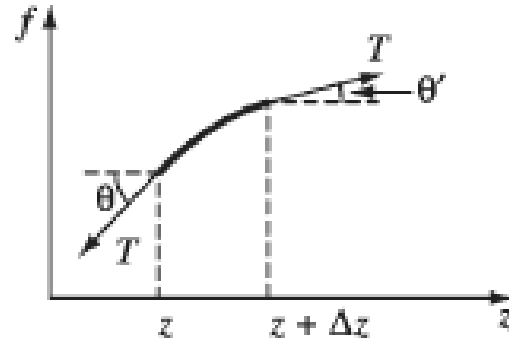
The displacement at point z , at the later *time* t , is the same as the displacement a distance vt to the left (i.e. at $z - vt$), back at *time* $t = 0$:

$$f(z, t) = f(z - vt, 0) = g(z - vt).$$

Electromagnetic Waves

Electromagnetic Waves

The Wave Equation for string



The net transverse force

$$\Delta F = T \sin \theta' - T \sin \theta,$$

$$\Delta F \cong T(\tan \theta' - \tan \theta) = T \left(\left. \frac{\partial f}{\partial z} \right|_{z+\Delta z} - \left. \frac{\partial f}{\partial z} \right|_z \right) \cong T \frac{\partial^2 f}{\partial z^2} \Delta z.$$

If the mass per unit length is μ , Newton's second law says

$$\Delta F = \mu(\Delta z) \frac{\partial^2 f}{\partial t^2},$$

$$\frac{\partial^2 f}{\partial z^2} = \frac{\mu}{T} \frac{\partial^2 f}{\partial t^2}.$$

$$\boxed{\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}}, \quad v = \sqrt{\frac{T}{\mu}}.$$

The wave **equation**, admits solutions of the form

$$f(z, t) = g(z - vt), \quad u \equiv z - vt$$

$$\frac{\partial f}{\partial z} = \frac{dg}{du} \frac{\partial u}{\partial z} = \frac{dg}{du}, \quad \frac{\partial^2 f}{\partial z^2} = \frac{\partial}{\partial z} \left(\frac{dg}{du} \right) = \frac{d^2 g}{du^2} \frac{\partial u}{\partial z} = \frac{d^2 g}{du^2},$$

$$\frac{\partial f}{\partial t} = \frac{dg}{du} \frac{\partial u}{\partial t} = -v \frac{dg}{du}, \quad \frac{\partial^2 f}{\partial t^2} = -v \frac{\partial}{\partial t} \left(\frac{dg}{du} \right) = -v \frac{d^2 g}{du^2} \frac{\partial u}{\partial t} = v^2 \frac{d^2 g}{du^2},$$

$$\frac{d^2 g}{du^2} = \frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 f}{\partial t^2}.$$

Also of the form

$$f(z, t) = h(z + vt).$$



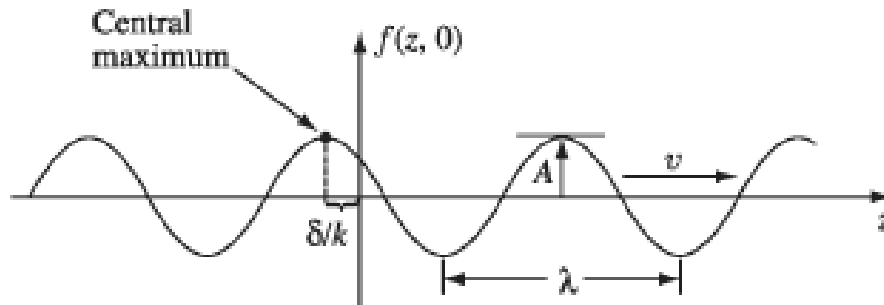
$$f(z, t) = g(z - vt) + h(z + vt).$$

Sinusoidal Waves

$$f(z, t) = A \cos[k(z - vt) + \delta]$$

Amplitude

phase



k = Wave number $\lambda = \frac{2\pi}{k}$

Period $T = \frac{2\pi}{kv}$

Frequency $\nu = \frac{1}{T} = \frac{kv}{2\pi} = \frac{v}{\lambda}$

Angular frequency $\omega = 2\pi\nu = kv$

$$f(z, t) = A \cos(kz - \omega t + \delta)$$

$$f(z, t) = \text{Re} [Ae^{i(kz - \omega t + \delta)}]$$

ELECTROMAGNETIC WAVES IN VACUUM

$$\left. \begin{array}{ll} \text{(i)} & \nabla \cdot \mathbf{E} = 0, \\ \text{(ii)} & \nabla \cdot \mathbf{B} = 0, \end{array} \right\} \quad \left. \begin{array}{ll} \text{(iii)} & \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \text{(iv)} & \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \end{array} \right\}$$
$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = \nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t} \right)$$
$$= -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2},$$


$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\left. \begin{array}{l} \nabla \times (\nabla \times \mathbf{B}) = \nabla(\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B} = \nabla \times \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \\ = \mu_0 \epsilon_0 \frac{\partial}{\partial t}(\nabla \times \mathbf{E}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}. \end{array} \right\}$$
$$\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}.$$

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad \nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}.$$

ELECTROMAGNETIC WAVES IN VACUUM

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

Let us consider the propagation along z-direction

$$\tilde{\mathbf{E}}(z, t) = \tilde{\mathbf{E}}_0 e^{i(kz - \omega t)}$$

Where $\tilde{\mathbf{E}}_0$ is complex amplitude (constant)

$$\nabla \cdot \mathbf{E} = 0 \quad \Rightarrow \quad \frac{\partial(E_{0x} e^{i(kz - \omega t)})}{\partial x} + \frac{\partial(E_{0y} e^{i(kz - \omega t)})}{\partial y} + \frac{\partial(E_{0z} e^{i(kz - \omega t)})}{\partial z} = 0$$

$$\Rightarrow E_{0z} = 0 \quad \text{No amplitude of oscillation along the propagation direction (transvers in nature)}$$

Similarly for $\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}$ if the propagation along z-direction, ie $\tilde{\mathbf{B}}(z, t) = \tilde{\mathbf{B}}_0 e^{i(kz - \omega t)}$

We can show $B_{0z} = 0$

ELECTROMAGNETIC WAVES IN VACUUM

From Faraday's law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\tilde{\mathbf{E}}(z, t) = \tilde{\mathbf{E}}_0 e^{i(kz - \omega t)}$$

$$\tilde{\mathbf{B}}(z, t) = \tilde{\mathbf{B}}_0 e^{i(kz - \omega t)}$$

$$\begin{aligned} LHS &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & 0 \end{vmatrix} = \hat{i} \left(-\frac{\partial E_y}{\partial z} \right) + \hat{j} \left(\frac{\partial E_x}{\partial z} \right) + \hat{k} \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \\ &= -ikE_{0y}\hat{i} + ikE_{0x}\hat{j} \end{aligned}$$

$$\begin{aligned} RHS &= -\frac{\partial \mathbf{B}}{\partial t} = -\hat{i} \left(\frac{\partial B_x}{\partial t} \right) - \hat{j} \left(\frac{\partial B_y}{\partial t} \right) - \hat{k} \cdot 0 \\ &= i\omega B_{0x}\hat{i} + i\omega B_{0y}\hat{j} \end{aligned}$$

➡ $kE_{0y} = -\omega B_{0x} \quad \& \quad kE_{0x} = \omega B_{0y}$

➡ $\omega \tilde{B}_0 = k(\hat{k} \times \tilde{E}_0)$

$$\hat{k} \times \tilde{E}_0 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ E_{0x} & E_{0y} & E_{0z} \end{vmatrix} = -E_{0y}\hat{i} + E_{0x}\hat{j}$$

ELECTROMAGNETIC WAVES IN VACUUM

From Faraday's law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\tilde{\mathbf{E}}(z, t) = \tilde{\mathbf{E}}_0 e^{i(kz - \omega t)}$$

$$\tilde{\mathbf{B}}(z, t) = \tilde{\mathbf{B}}_0 e^{i(kz - \omega t)}$$

$$\begin{aligned} LHS &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & 0 \end{vmatrix} = \hat{i} \left(-\frac{\partial E_y}{\partial z} \right) + \hat{j} \left(\frac{\partial E_x}{\partial z} \right) + \hat{k} \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \\ &= -ikE_{0y}\hat{i} + ikE_{0x}\hat{j} \end{aligned}$$

$$\begin{aligned} RHS &= -\frac{\partial \mathbf{B}}{\partial t} = -\hat{i} \left(\frac{\partial B_x}{\partial t} \right) - \hat{j} \left(\frac{\partial B_y}{\partial t} \right) - \hat{k} \cdot 0 \\ &= i\omega B_{0x}\hat{i} + i\omega B_{0y}\hat{j} \end{aligned}$$

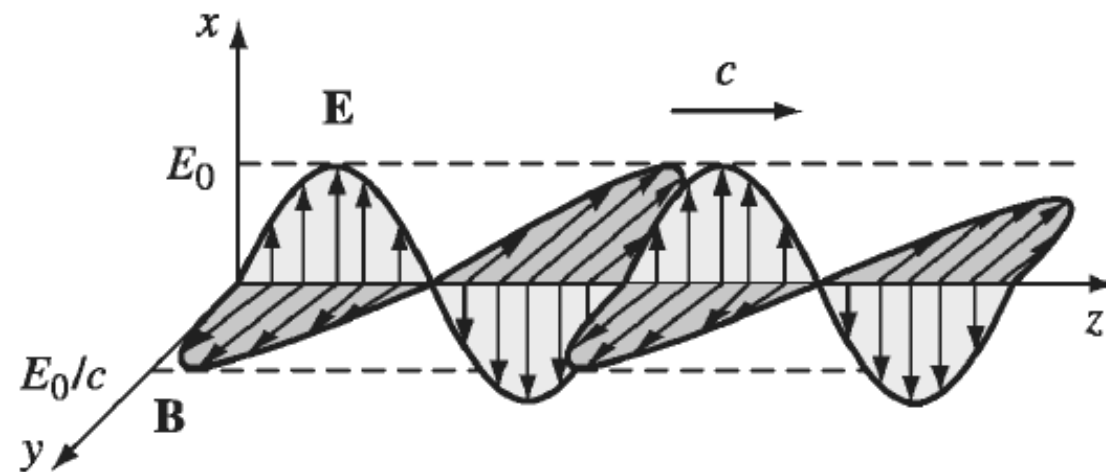
$$\Rightarrow kE_{0y} = -\omega B_{0x} \quad \& \quad kE_{0x} = \omega B_{0y}$$

$$\Rightarrow \omega \tilde{\mathbf{B}}_0 = k(\hat{z} \times \tilde{\mathbf{E}}_0)$$

Both $\vec{E}$ and $\vec{B}$ are in the same phase
and are mutually perpendicular

$$\omega = ck \Rightarrow E_0 = cB_0$$

ELECTROMAGNETIC WAVES IN VACUUM



ELECTROMAGNETIC WAVES IN VACUUM

From Faraday's law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\tilde{\mathbf{E}}(z, t) = \tilde{\mathbf{E}}_0 e^{i(kz - \omega t)} \quad \Rightarrow \quad \tilde{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \hat{\mathbf{n}}$$

$$\tilde{\mathbf{B}}(z, t) = \tilde{\mathbf{B}}_0 e^{i(kz - \omega t)} = \frac{1}{c} \tilde{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} (\hat{\mathbf{k}} \times \hat{\mathbf{n}}) = \frac{1}{c} \hat{\mathbf{k}} \times \tilde{\mathbf{E}},$$

- $\hat{\mathbf{n}}$ is the polarization vector (along $\vec{E}$)
- $\hat{\mathbf{k}}$ is the propagation or wave vector pointing in the direction of propagation

Electromagnetic wave

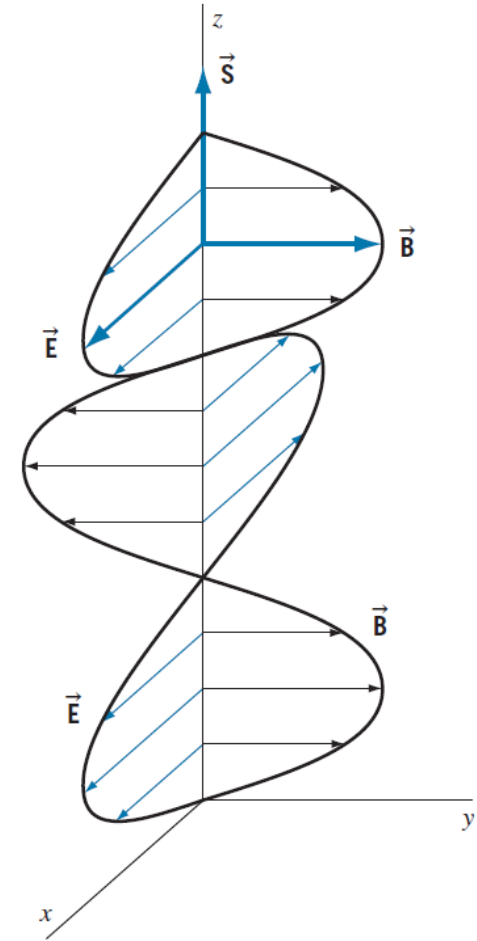
$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}$$

$$\vec{\mathbf{E}} = \vec{\mathbf{E}}_0 \sin(kz - \omega t), \quad \vec{\mathbf{B}} = \vec{\mathbf{B}}_0 \sin(kz - \omega t) \quad B_0 = \frac{E_0}{c}$$

Poynting vector :

$$\vec{\mathbf{S}} = \frac{1}{\mu_0} \vec{\mathbf{E}} \times \vec{\mathbf{B}}$$

$$\begin{aligned} \vec{\mathbf{S}} &= \frac{1}{\mu_0} E_0 B_0 \sin^2(kz - \omega t) \hat{\mathbf{k}} \\ &= \frac{1}{\mu_0 c} E_0^2 \sin^2(kz - \omega t) \hat{\mathbf{k}} \end{aligned}$$



The Poynting vector has dimensions of power (energy per unit time) per unit area

Electromagnetic wave

The power propagates along z direction is

$$P = \frac{1}{\mu_0 c} E_0^2 A \sin^2(kz - \omega t)$$

Energy carried through wave-front

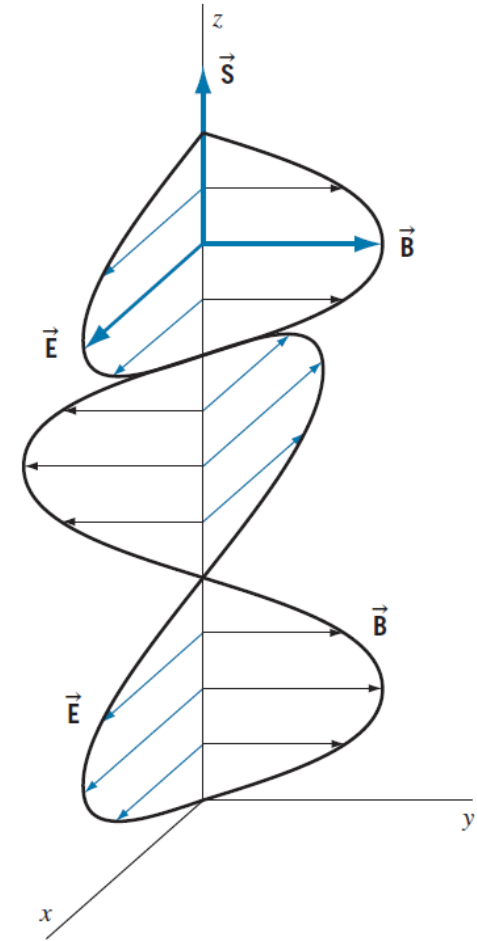
Important NOTE:

- The power fluctuates with time with a high frequency $2(\omega/2\pi) \approx 10^{15}$

We observe an average value $P_{\text{av}} = \frac{1}{T} \int_0^T P dt$

- The intensity (the average power per unit area) is proportional to E_0^2 . This is a general property of waves: *the intensity is proportional to the square of the amplitude.*

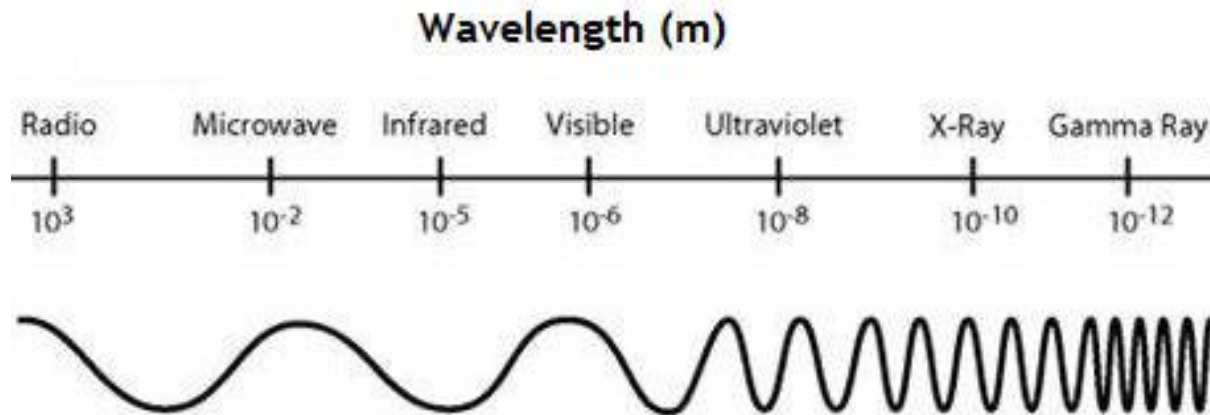
$$I = \frac{P_{\text{av}}}{A} = \frac{1}{2\mu_0 c} E_0^2$$



Electromagnetic wave

Velocity of the electromagnetic radiation is constant in free space

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \lambda \cdot \nu \cong 3 \times 10^8 \text{ m/s}$$



Using the wave nature of light was difficult to explain the nature of the black body radiation spectrum !!!

Curl of the Electric Field:

Let's take the simplest electric field:

Electric field due to a single point charge q is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}$$

We need to find the curl of it

$$\nabla \times \mathbf{E}(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \nabla \times \left(\frac{1}{r^2} \hat{\mathbf{r}} \right)$$

Take the area integral

$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \int_{surf} \nabla \times \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) \cdot d\mathbf{a}$$

Use Stokes's theorem

$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \oint \left(\frac{1}{r^2} \hat{\mathbf{r}} \right) \cdot d\mathbf{l} = \frac{q}{4\pi\epsilon_0} \oint \frac{1}{r^2} dr$$

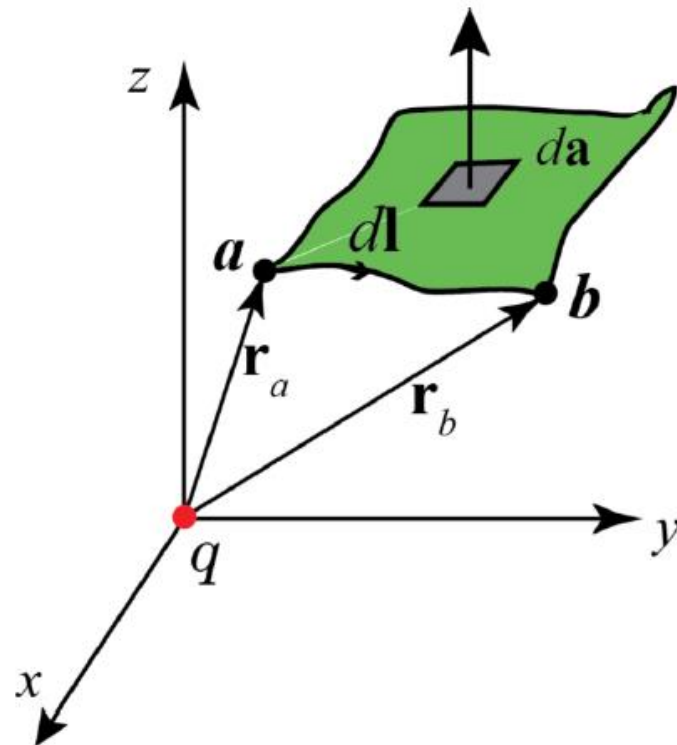
since

$$d\mathbf{l} = dr\hat{\mathbf{r}} + rd\theta\hat{\boldsymbol{\theta}} + r\sin\theta d\phi\hat{\boldsymbol{\phi}}$$

$$\int_{surf} \nabla \times \mathbf{E}(\mathbf{r}) \cdot d\mathbf{a} = \frac{q}{4\pi\epsilon_0} \times \frac{1}{r} \Big|_{r_a}^{r_a} = 0$$

implies

$$\nabla \times \mathbf{E} = 0$$



Electric Potential:

Recall: If the curl of a vector field $\mathbf{F}$ is zero, that is, if $\nabla \times \mathbf{F} = 0$ everywhere, then:

(1) $\int_a^b \mathbf{F} \cdot d\mathbf{l}$ is independent of path.

(2) $\oint \mathbf{F} \cdot d\mathbf{l} = 0$ for any closed loop.

(3) $\mathbf{F}$ is the gradient of a scalar function: $\mathbf{F} = -\nabla V$

This is because of Stokes' theorem

$$\int_{\text{Surf}} (\nabla \times \mathbf{F}) \cdot d\mathbf{a} = \oint_{\text{Path}} \mathbf{F} \cdot d\mathbf{l}$$

• This is because Curl of a gradient is zero $\nabla \times (\nabla V) = 0$

The curl of Electric field $\mathbf{E}$ is zero, that is, $\nabla \times \mathbf{E} = 0$ everywhere. Therefore:

$\oint \mathbf{E} \cdot d\mathbf{l} = 0$ for any closed loop.



$\int_a^b \mathbf{E} \cdot d\mathbf{l}$ is independent of path.

$\mathbf{E}$ is the gradient of a scalar function:

$$\mathbf{E} = -\nabla V$$

V is called the electric potential. It is a scalar quantity, the gradient of which is equal to the electric field

How to obtain/write the electric potential in terms of $\mathbf{E}$??

Electric Potential:

Since $\nabla \times \mathbf{E} = 0$ everywhere, implies $\int_a^b \mathbf{E} \cdot d\mathbf{l}$ is independent of path(depends only on the end points)

$$\int_a^b \mathbf{E} \cdot d\mathbf{l} = V(a) - V(b)$$

Use the fundamental theorem of gradient : $\int_a^b \nabla V \cdot d\mathbf{l} = V(b) - V(a)$

$$\int_a^b \nabla V \cdot d\mathbf{l} = - \int_a^b \mathbf{E} \cdot d\mathbf{l}$$

$$\mathbf{E} = -\nabla V$$

Ex1. Two identical uniform line charges lie along the x and y axis with densities $\rho_l = 20 \mu\text{C}/\text{m}$. Obtain the electric field at a point (3m,3m,3m)

Ex2. Two infinite uniform sheets of charge each with density ρ_s are located at $x = \pm 1$. Determine the electric field in all regions.